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# Early Proterozoic palaeomagnetism: new results from the intrusives and related rocks of the Karelian, Belomorian and Kola provinces, eastern Fennoscandian Shield

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## SUMMARY

The palaeomagnetic data that are available for the eastern Fennoscandian Shield are too sparse to allow any palaeomagnetic constraints to be determined for intracratonic horizontal movements in this part of the shield. To address this issue we have obtained new palaeomagnetic data from mafic and ultramafic plutons, sills, dykes and basement country rocks in the Kola, Belomorian and Karelian provinces of the Fennoscandian Shield.

A number of natural remanent magnetization (NRM) components were isolated, and we associate these with various events of Precambrian tectonomagmatic activity within the eastern Fennoscandian Shield. 36 palaeomagnetic poles were obtained. 12 poles, with ages ranging from 1700 Ma to 2450 Ma, satisfy the accepted reliability criteria (corresponding to the 'B-category' of Briden & Duff 1981). Of these, six might be classified as reference poles. For the Kola and Belomorian provinces there are poles from the Main Range gabbro-dolerites (16.7°S, 304.8°E, $A_{95} = 8.3°$; U–Pb age of $2436 \pm 25$ Ma), the Voche-Lambina ultramafics and amphibolites (12.9°N, 314.1°E, $A_{95} = 8.6°$; Rb–Sr age of $2198 \pm 172$ Ma), and the Kolvitsa peridotites and schists (44.2°N, 232.6°E, $A_{95} = 5.4°$; assigned magnetization age of 1900 Ma). For the Karelian Province, reference poles are presented for the Burakovka gabbro-norites (19.5°S, 262.6°E, $A_{95} = 7.2°$; Sm–Nd age of $2439 \pm 22$ Ma), the Matozero doleritic dykes (20.6°S, 308.5°E, $A_{95} = 12.3°$; U–Pb age of $2150 \pm 60$ Ma), and the Ropruchey Sill gabbro-dolerites (40.5°N, 229.8°E, $A_{95} = 8.1°$; U–Pb age of $1770 \pm 12$ Ma). Although some NRM components of other rocks are not as precisely dated, a number of palaeomagnetic poles are in good internal agreement, forming two tracks of the Lower Proterozoic apparent polar wander path (APWP). A preliminary version of the pre-Svecofennian APWP tracks for the Karelian and Kola–Belomorian provinces is proposed. Our results confirm the general drift pattern of Fennoscandia in the Proterozoic and support the idea of the absence of large-scale block movements within the Fennoscandian Shield since Jatulian times (2300–2100 Ma). To help constrain the relative positions of the various tectonic blocks of the eastern Fennoscandian Shield during pre-Jatulian times, more new palaeomagnetic and radiometric data need to be added to the available palaeomagnetic database.

**Key words:** APW (apparent polar wander), Early Proterozoic, Fennoscandian Shield, palaeomagnetism.

## 1 INTRODUCTION

Recently, a series of global and large-scale palaeomagnetic reconstructions for Precambrian time have been presented (Powell *et al.* 1993; Torsvik *et al.* 1996). They demonstrate the history of horizontal movements of the Precambrian cratons, and the creation and destruction of large continental assemblies such as Ur, Nena and Rodinia (e.g. Rogers 1996). The Precambrian cratons are considered as coherent units in most of these reconstructions. However, the determination of the

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extent of the relative horizontal movements of the various lithospheric blocks that make up the Precambrian cratons, the time of their origin and mode of their consolidation (Gaal & Gorbatschev 1987) is of special importance in understanding the nature and style of plate tectonics in the Precambrian. Palaeomagnetic data offer the possibility of addressing some of these questions; until now, however, only the Canadian Shield has yielded enough Precambrian palaeomagnetic data to indicate significant intracratonic movements (Symons 1991). As for the Fennoscandian Shield, there is as yet no palaeomagnetic evidence that can resolve that degree of movement.

An apparent polar wander path (APWP) for the Fennoscandian Shield has recently been constructed for the time interval 2850–600 Ma, under the assumption that the Shield remained tectonically coherent during this time (Elming *et al.* 1993). There is, however, geological evidence which indicates that significant intracratonic occurred during this time. Gorbatschev & Bogdanova (1993) provided evidence for rifting and the possible break-up of the shield between 2500 and 2000 Ma. The resultant basins were then closed due to a collisional orogeny between 1950 and 1820 Ma, possibly marking the formation of a Proterozoic supercontinent (Piper 1987). On the other hand, the Archaean and Palaeoproterozoic palaeomagnetic data from the Fennoscandian Shield were predominantly obtained from the Karelian and Svecofennian provinces. Few poles from the Lapland granulite belt of the Belomorian Province and from Kola Province were available for the Elming *et al.* (1993) compilation and their reconstruction of the Fennoscandian APWP. Furthermore, the database used for this purpose (Pesonen *et al.* 1991; Elming *et al.* 1996) contains very few reliable pre-Svecofennian (>1850 Ma) palaeomagnetic poles.

Recently, a series of studies on the eastern part of the shield has been carried out by various workers (Krasnova & Gooskova 1995; Mertanen 1995; Torsvik & Meert 1995; Dudkin & Turemnov 1995; Goriainov & Turemnov 1996; Damm *et al.* 1997; Khramov *et al.* 1997). These have enlarged the Precambrian palaeomagnetic database for Fennoscandia, and this study further updates this database.

We present new palaeomagnetic results from mafic and ultramafic intrusions and associated rocks of the Karelian, Belomorian and Kola provinces of the Fennoscandian Shield. We will use these data to test the post-Archaean coherence of the shield megablocks (Pesonen & Neuvonen 1981) and to obtain more reliable estimates on the kinematics of the various shield units, in particular for Jatulian times (2300–2100 Ma).

## 2 GEOLOGICAL SETTING AND SAMPLING

The eastern part of the Fennoscandian Shield is divided into four provinces, each different in age, as well as having been formed by different tectonic, magmatic and metamorphic processes (Sokolov 1987). These units are shown in Fig. 1; their names, from north to south, are the Kola, Belomorian, Karelian and Ladoga blocks, extending as far west as Finland, and they are overlain by Phanerozoic sediments of the Russian Plate to the southeast (Sokolov 1987). These units form two domains, representing the first stage of the Fennoscandian continental lithospheric growth, with the Kola and Belomorian units forming the Archaean domain, and the Karelian and Ladoga units, the Archaean–Early Proterozoic domain.

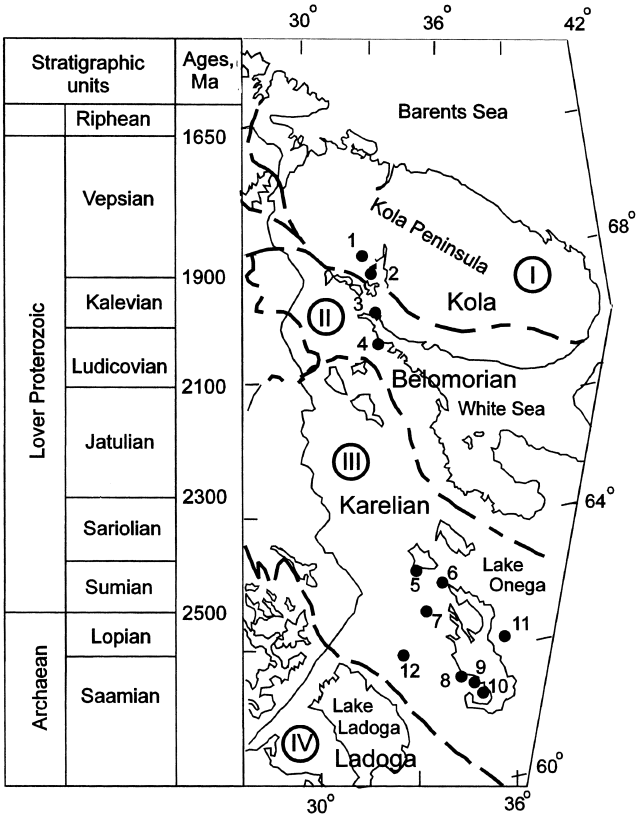



**Figure 1.** Tectonic and stratigraphic units of the eastern Fennoscandian Shield and areas of palaeomagnetic study—1: Moncha, 2: Voche–Lambina, 3: Kolvitsa, 4: Tolstik, 5: Segozero, 6: Matozero, 7: Girvas, 8: Ladva, 9: Shoksha, 10: Sheltozero, 11: Burakovka, 12: Hautavaara. Tectonic blocks provinces—I: Kola, II: Belomorian, III: Karelian, IV: Ladoga (simplified after Gaal & Gorbatschev 1987). Stratigraphic units are drawn according to scheme of Semikhatov *et al.* (1991).

The major part of the Precambrian crust in this region was reworked during Early Proterozoic orogenic events. The degree of this reworking varies: it was intense and widespread in the Archaean domain within the Kola and Belomorian units, while in the Karelian unit the Proterozoic reworking occurred only within the rift zones that show mafic activity (Gaal & Gorbatschev 1987).

Our palaeomagnetic studies include the Lower Proterozoic Karelian Complex of the Karelian unit, and coeval plutonic and country metamorphic rocks of the Belomorian and Kola units. The stratigraphic scheme of the formations sampled is depicted in Fig. 1.

In the Kola block the palaeomagnetic results were obtained from the Monchegorsk ultramafic intrusion and related rocks of Main Range in the Moncha area, and from the metamorphic-intrusive complex in the Voche–Lambina area. A similar complex was examined in the Kolvitsa and Tolstik areas of the Belomorian block (Fig. 1). The Jatulian and Svecofennian (Vepsian) formations were studied in the Onega district in South Karelia. They comprise dolerites, gabbro-dolerites and schists from the Segozero, Girvas, Matozero, Shoksha, Sheltozero and Ladva areas. The Burakovka layered intrusion of the Trans-Onega area and the Hautavaara greenstone structure of southwestern Karelia were also sampled (Fig. 1).

We focus here on those new data which have radiometric control, field tests and rock-magnetic results. Other new data, which we have recently published (Khramov *et al.* 1997) and updated, will be presented in a later section in summary form.

The Monchegorsk ultramafic layered pluton is situated in the central part of the Kola Peninsula, to the north of the Pechenga–Varzuga Tectonic Zone (Fig. 1). It is made up of two limbs (Fig. 2). The northwestern limb consists mainly of ultramafic rocks (dunites, harzburgites and pyroxenites), whereas the eastern limb exposes mafic-ultramafic rocks of the upper part of the intrusion (Balashov *et al.* 1993). Zircons separated from gabbro-norites of this limb yield a U–Pb age of $2493 \pm 7$ Ma ($2\sigma$ errors are quoted in all cases). The pluton is cut by picritic dykes, which are, in turn, cut by two generations of metadoleritic dykes. The ages of these dykes are still being debated, but are probably within the range 2200–1800 Ma (Gorbunov *et al.* 1985).

The gabbro-doleritic Main Range massif extends southwards from the Monchegorsk pluton. These Main Range rocks have yielded a U–Pb zircon isochron age of $2436 \pm 25$ Ma (Bogdanova & Bibikova 1993). Palaeomagnetic sampling was carried out in all the structures mentioned above (Fig. 2).

The Voche–Lambina area is located west of Lake Imandra, along the line of the Pechenga–Varzuga Tectonic Zone (Fig. 1). The area has been described as the contact zone between 'Saamian' orthogneisses and Lopian supracrustal rocks representing a greenstone belt (Daly *et al.* 1993). Lopian rocks form a sequence of metamorphosed acid, intermediate and mafic volcanics, and clastic sediments that tectonically overlie 'Saamian' gneisses. The supracrustals have been divided into four 'formations'; however, all of the contacts between these are interpreted as tectonic (Fig. 2). The metamorphic rocks exhibit an amphibolite facies alteration (approximately 610–620 °C), but locally the rocks have experienced younger epidote-amphibolite facies metamorphism (430–520 °C). The youngest doleritic dykes, which cut all these rocks, show a faint auto-metamorphism. These dykes are mostly of Svecofennian (1880–1810 Ma) age (Gorbunov *et al.* 1985). Many of the ages for metamorphic rocks have been interpreted as secondary, especially Rb–Sr isochrons. These data have been summarized by Mitrofanov *et al.* (1991), with the oldest gneisses yielding a Rb–Sr age of $2540 \pm 46$ Ma. These gneisses are cut by granodiorites which gave a Rb–Sr age of $2198 \pm 172$ Ma. The underlying 'Saamian' gneisses yielded a U–Pb zircon age of $2801 \pm 9$ Ma, indicating a Late Archaean (Lopian) age for these gneisses. A Lopian rhyodacite from formation IV (Fig. 2) has yielded a U–Pb zircon age of $2800 \pm 150$ Ma and a Rb–Sr age of $2450 \pm 83$ Ma (Mitrofanov *et al.* 1991). We have sampled the dolerites from the dykes and country gneisses, amphibolites and ultramafics (Fig. 2).

The Kolvitsa area is located on both sides of the mouth of the river Kolvitsa on the northeast shore of Kandalaksha Bay (Fig. 2). This area lies in the Lapland deep fault zone, which separates the Kola and Belomorian blocks. Crystalline schists and gneisses, with a U–Pb zircon age of $2620 \pm 25$ Ma (A. B. Vrevsky, personal communication, 1991) are intruded by a gabbro-anorthosite layered intrusion, with a U–Pb zircon age of $2450 \pm 10$ Ma (Mitrofanov *et al.* 1993), and by submeridional mafic-ultramafic porphyritic dykes. Amphibolitization of the gabbros indicates that the first metamorphic event occurred at $2423 \pm 3$ Ma (U–Pb zircon age; Mitrofanov *et al.*

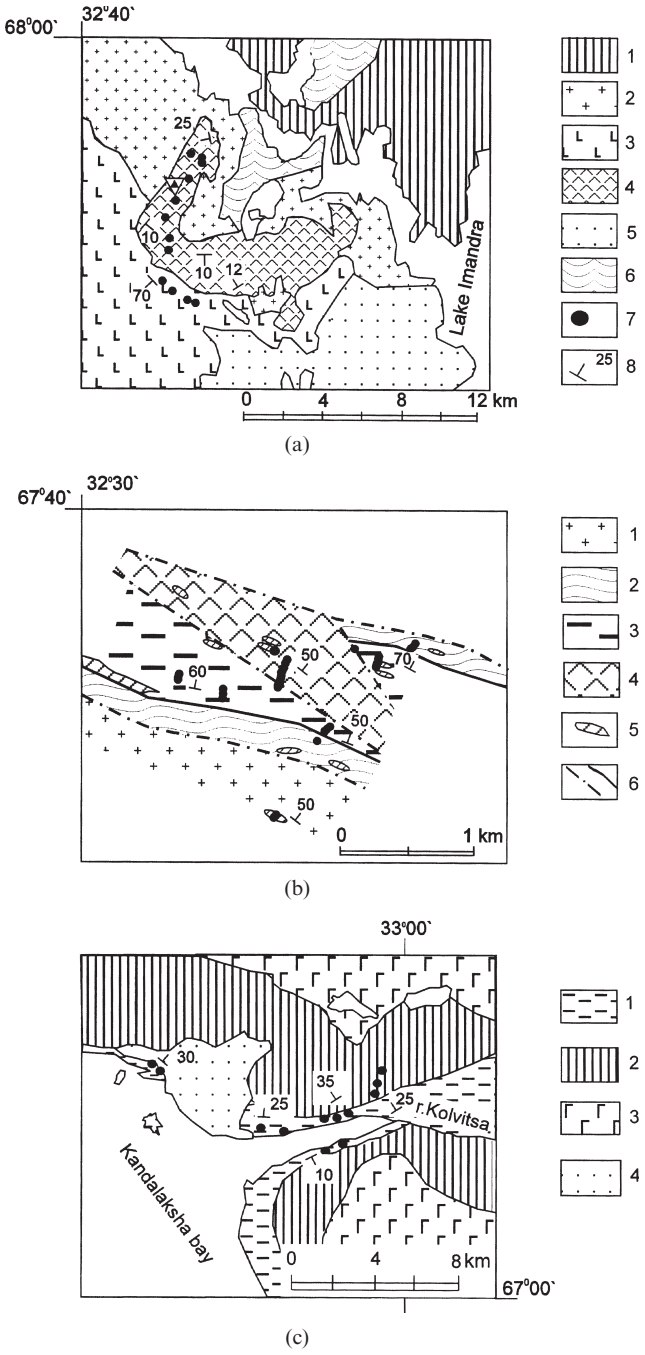



**Figure 2.** Geological sketch maps of the palaeomagnetic sampling localities, Kola and Belomorian provinces. Simplified from geological maps in Goriainov (1971), Guidebook (1984), Bogdanova & Bibikova (1993). (a) Moncha area. 1: Archaean gneisses, 2: Upper Archaean–Lower Proterozoic ferriferous quartzites, 3: Main Range gabbro-anorthosites, 4: Monchegorsk ultramafic intrusion, 5: Imandra-Varzuga formation rocks, 6: Quaternary cover. For all maps—7: sampling localities, 8: strike and dip of bedding or foliation. (b) Voche–Lambina area. 1: Archaean granites and gneisses, 2: acid metavolcanites, 3: volcano-clastic member, 4: rhyolite-dacite member, 5: mafic and ultramafic intrusions, 6: discordant and concordant geological boundaries. (c) Kolvitsa area. 1: gneisses and amphibolites of Khetolambina Fm, 2: amphibolites and amphibolized gabbro, 3: gabbro-anorthosites, 4: Quaternary sediments.

1993). Samples were taken from these dykes as well as from the country-rock schists.

Areas in the Onega Depression, which runs southwards into the West Onega Trough (Fig. 3), have also been studied. This depression is composed of 3 km of Lower Proterozoic strata, comprising low-grade sedimentary rocks and lavas, intruded by numerous doleritic dykes and sills. Three phases of Jatulian (*ca.* 2300–2100 Ma) volcanism are recorded: the first, and oldest, is represented by basaltic lavas; the second phase, by subvolcanic sills; and the third phase, by dykes and intrusive stocks. The U–Pb zircon age of dykes gives a mean value of $2150 \pm 60$ Ma, whilst the oldest age of basalts is constrained by a $2420 \pm 45$ Ma U–Pb zircon age of pebbles in the basal conglomerates (Sokolov 1987). Doleritic sills, which intrude Upper Jatulian dolomites (with a Pb–Pb age determined from authigenic minerals of $2300 \pm 140$ Ma), yield the youngest U–Pb zircon age of 2070 Ma (Lobach-Zhuchenko *et al.* 1986).

In this study we focus primarily on the Matozero area, sampled more carefully than the other areas investigated (Segozero and Girvas; Fig. 3). The Matozero area is situated near Lake Matozero, west of the town of Medvezhiegorsk at the northern termination of Povenetsky Bay of Lake Onega. The main feature of this area is the presence of the Medvezhiegorsk Sill, which is 16 km long and 150 m thick. The lower part of this sill is composed of dolerites, whereas its upper part is composed of leucodolerites with high concentrations of albite. The subhorizontal sill intrudes into flat-lying Upper Jatulian (2200–2100 Ma) schists of the east–west-trending Kuulina Zone. The sill and country rocks were subjected to low-grade greenstone metamorphism. Dolerites from the sill, and baked and country schists have been sampled for palaeomagnetic study.

The West Onega Trough is composed of Vepsian-aged flat-lying grey sandstones and siltstones of the Petrozavodsk

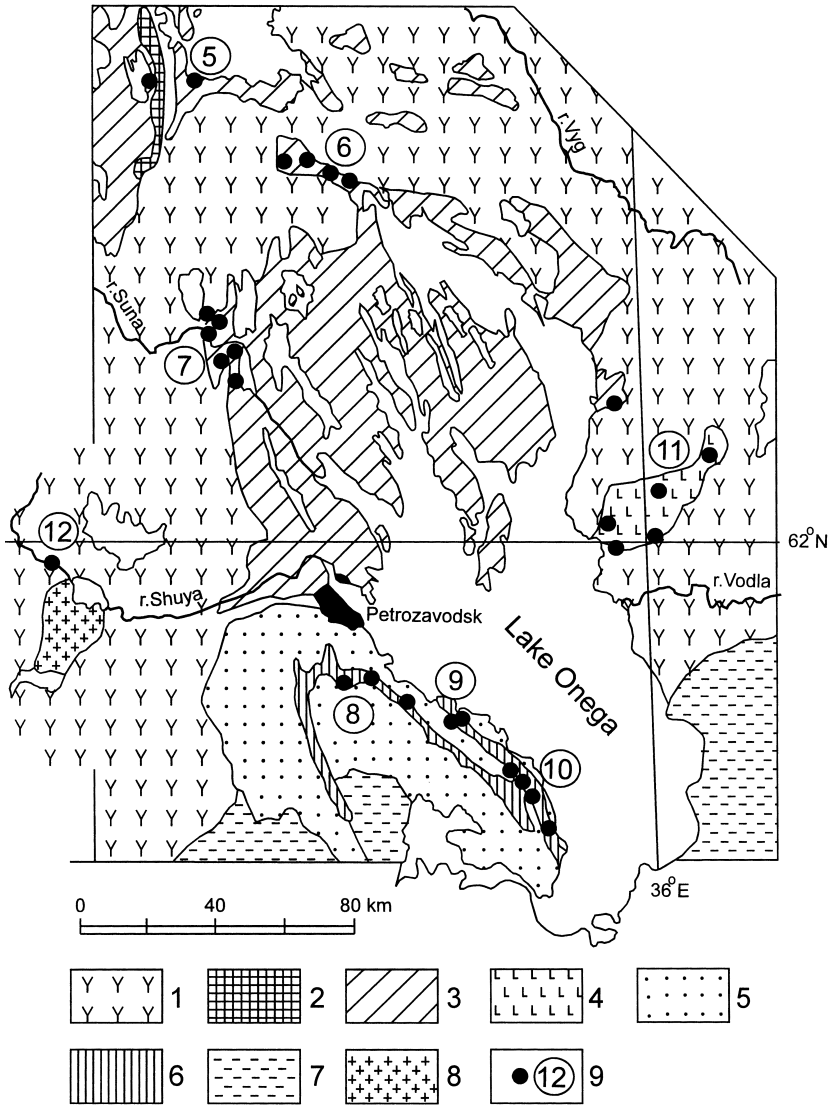



**Figure 3.** Geological map of the Onega region and palaeomagnetic sampling areas, numbered as in Fig. 1: simplified from Atlas Karelskoi ASSR (1989). 5: Segozero, 6: Matozero, 7: Girvas, 8: Ladva, 9: Shoksha, 10: Sheltozero, 11: Burakovka, 12: Hautavaara. Legend—1: Archaean granitoides, 2: Sariolian and Sumian schists and volcanites, 3: Jatulian and Suisarian quartzites, dolomites, schists and mafic volcanites, 4: mafic and ultramafic rocks of the Burakovka layered intrusion, 5: quartzites of the Shoksha formation, 6: gabbro-dolerites of the Ropruchey Sill, 7: Vendian to Devonian sedimentary cover, 8: rapakivi granites, 9: sampling localities.

Formation and the overlying red quartzites of the Shoksha Formation; near the formation margin, two basaltic flows were observed. The strata are intruded by the Ropruchey Sill. This layered lopolith-like sill is some 170 m thick and covers an area of about 4500 km$^2$. The outcrops of the sill form two wings: the western outcrops in the Ladva area consist mainly of gabbro and gabbroic syenites from the central layers of the sill, whereas the eastern outcrops in the Shoksha and Sheltozero areas are made up of dolerites from the bottom part of the sill. These dolerites are cut by monzonite veins, related to the last stage of sill emplacement. Similar veins are also observed in the quartzites of the Shoksha Formation. In the Rybreka area (not sampled here) located near the Sheltozero area, zircons from the monzonites yield a U–Pb age of 1770 $\pm$ 12 Ma (Bibikova *et al.* 1990). The U–Pb age of the authigenic minerals from the Petrozavodsk Formation is 1810 $\pm$ 100 Ma, while the K–Ar age of the Ropruchey Sill is 1690–1640 Ma (Semikhatov 1974). Samples from this sill were taken in the Sheltozero, Shoksha and Ladva areas (Fig. 3).

The Burakovka mafic-ultramafic intrusion, which cuts the Archean and Lower Proterozoic gneisses and granites, was examined in the eastern Trans-Onega region (Fig. 3). The intrusion covers an area of about 800 km$^2$ and is associated with a deep fault. The massif consists of three blocks: the Burakovka, Central and Aganozero blocks. Its cross-sectional form is asymmetrical and cup-like, with the magma feeder within the Aganozero block. The massif is layered. Its lower part contains olivinites and peridotites, the middle part exposes gabbro-norite horizons, and the upper part, besides gabbro-norites, contains a small amount of gabbro-diorites. The Sm–Nd age of the Burakovka intrusion is 2439 $\pm$ 22 Ma (Amelin & Semenov 1990), while the U–Pb zircon age of its gabbro-norites is 2449 $\pm$ 1 Ma (Amelin *et al.* 1995). All three blocks of the Burakovka intrusion were sampled for the purpose of palaeomagnetic investigation, including the gabbro-dolerite dykes framing the intrusion in the southwest which form offshoots of the main intrusion.

The Hautavaara area is about 80 km west of Petrozavodsk in the valley of the river Shuya, near Lake Shotozero (Figs 1 and 4). This area is situated on the southern termination of the Vedlozero–Segozero greenstone belt, near the northern boundary of the Ulalega Massif of rapakivi granites which yield a U–Pb zircon age of 1543 $\pm$ 8 Ma (Neymark *et al.* 1994). The main outcrops in the area are Archaean acid rocks of the Ignoila volcano. These rocks are represented by andesitic and dacitic lavas, and the neck of dacite is cut by amphibitolized gabbroic dykes. These lavas yield a U–Pb zircon age of 2945 $\pm$ 19 Ma, whereas the dioritic massif, which cuts these lavas, yields a U–Pb zircon age of 2745 $\pm$ 62 Ma (Vrevsky *et al.* 1996). The U–Pb zircon systems for these rocks showed disturbances consistent with the timing of Svecofennian metamorphic events. These events are confirmed by K–Ar ages of 1890–1830 Ma for lavas and Pb–Pb model ages of 1900–1850 Ma for sulphides from the Hautavaara mineral deposit. Amphibolized gabbro and andesitic-dacitic lavas were sampled here (Fig. 4).

In the field from nine to 54 hand samples were taken from four to 18 sites in each sampling area. Magnetic and sun compasses were used for sample orientation.

## 3 LABORATORY PROCEDURES

Hand samples were cut into 24 $\times$ 24 $\times$ 24 mm cube-shaped specimens. Two cubes were prepared from each sample.

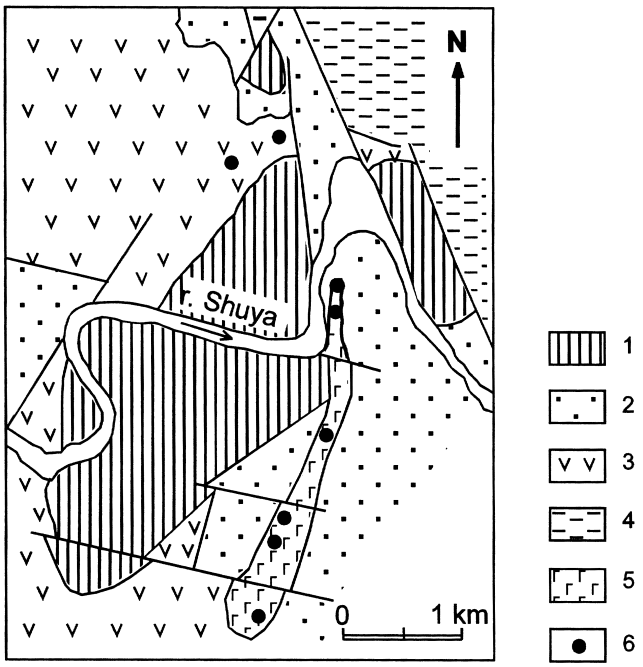



**Figure 4.** Geological sketch map of the Ignoila palaeovolcano, Hautavaara area. 1: dacitic lavas, 2: andesitic lavas, 3: agglomerates and tuffs, 4: schists, 5: gabbrioc amphibolized dykes, 6: sampling localities.

Measurements were carried out on astatic magnetometers (LAM-24; Brno production). Shielded furnaces with a residual magnetic field not exceeding 15 nT were used for stepwise thermal demagnetization. AF (alternating field) demagnetization was performed using a two-axis tumbling AF demagnetizer, housed in a three-component Helmholtz coil system. Components of NRM were identified by least-squares analysis (Kirschvink 1980) of linear segments of the demagnetization trajectories in orthogonal vector projections (Zijderveld 1967). The mean directions and associated statistical parameters (Fisher 1953) were calculated at the sample level. Reversal tests were carried out (McFadden & McElhinny 1990) and used to estimate the isolation quality of the dual-polarity NRM components and to detect possible age differences between normal and reversed magnetizations. Fold tests were used in the cases of non-horizontal layering or foliation of the intrusive and metamorphic rocks. Carriers of magnetization were studied for each group of rocks with certain directions of NRM. Thin and polished sections were investigated in transmitted and reflected light using an optical microscope and a microprobe. Thermomagnetic analyses were also carried out to determine the thermal dependence of saturated remanent magnetization $J_{rs}(T)$. The magnetic 'hardness' of the sampled rocks was estimated using the $Q_{rs}$ value, where $Q_{rs} = J_{rs}/kH$, where $k$ is the magnetic susceptibility and $H$ is a 0.05 mT magnetic field. The carriers of certain components of NRM were identified according to the unblocking temperature spectra ($T_{ub}$) and coercive spectra ($H_c$) of these components.

## 4 PALAEOMAGNETIC ANALYSIS

### 4.1 NRM components of rocks

During thermal demagnetization and AF demagnetization, two or three magnetization components were isolated for the

majority of specimens. These components have different ranges of unblocking temperatures and coercive spectra. Moreover, different rock types from the same site sometimes exhibit different characteristic remanent magnetization (ChRM) directions (Figs 5, 6, 7 and 8).

In the Moncha area, four groups of ChRM directions are isolated (Fig. 9). The first group (component A) is typical for the Main Range gabbro and for some pyroxenites of the Monchegorsk pluton (Figs 5 and 9a). It is a dual-polarity component with W-up/E-down directions. Dolerite dykes of the first generation and micropyroxenites at the dyke contacts carry a single-polarity $B_1$ component (Fig. 9c). The second generation of dykes, the so-called Kumuzhya dykes, which cut all rocks, carry the $B_2$ component (Figs 5 and 9e). A further component (C) (Figs 5 and 9d) is isolated in picrites of the Monchegorsk pluton and host gabbro. Components $B_2$ and C are also of single polarity here (Table 1).

$T_{ub}$ and $H_c$ values (the isolation ranges) of the NRM components show that magnetite is the main carrier of NRM, indicating a probable thermoremanent or chemical (crystallization) origin of these components (Fig. 5).

In the Voche–Lambina area, four groups of directions are isolated. They are similar to the directions of the Moncha groups. The first group (component A; Figs 6 and 9b) is of dual polarity with relatively high unblocking temperatures ($T_{ub} = 300$–$500\,^{\circ}C$) and is typical for the country ultramafics, gneisses and amphibolites. The second group of directions, component $B_1$ (Figs 6 and 9c), in contrast to the Moncha area, also display dual polarity (NW-up/SE-down), with lower unblocking temperatures ($T_{ub} = 100$–$450\,^{\circ}C$). Component $B_1$ is isolated in doleritic dykes, both in the Moncha and Voche–Lambina areas, whereas component C (Figs 6 and 9d), as well as component A (Figs 6 and 9b) are characteristic for the country rocks. The predominantly normal-polarity $B_2$ component is recorded in all rock types (Fig. 9f, Table 1).

In the Kolvitsa area, most specimens of the peridotitic dykes and basaltic porphyrites, as well as the country rock schists, contain two components of magnetization (Fig. 6). Bends in the Zijderveld diagrams, corresponding to the overlap of $T_{ub}$ spectra, are recorded at about 350–450 °C, with similar bends during AF demagnetization at 10–11 mT. The first low-$T_{ub}$ and low-coercivity component has a similar W-up direction to

**Table 1.** Mean remanence directions of rocks from the Kola and Belomorian provinces.

| Numbers Rock unit | NRM component | Isolation interval °C, mT | Sites/ /samples/ /specimens | $D^{(\circ)}$ | $I^{(\circ)}$ | K | $\alpha_{95}^{(\circ)}$ | S |
|---|---|---|---|---|---|---|---|---|
| **Moncha area, 68.0°N, 32.5°E** | | | | | | | | |
| 1.1 Monchegorsk pluton picrites and adjacent country rocks | C | 300–500 | 3/8/8 | 256.9 | 40.4 | 33.2 | 8.6* | 6.5 |
| 1.2 Monchegorsk pluton pyroxenites | A | 460–550 | 3/8/8 | 76.6 | 29.0 | 22.9 | 11.8* | 3.4 |
| 1.3 Gabbro-dolerites of the Main Range | A | 350–550, >40 | 4/10/10 | 261.5 | −27.5 | 19.4 | 11.3* | 3.3 |
| 1.4 Dolerite dykes of first generation and micropyroxenites | $B_1$ | 300–500, >100 | 2/6/9 | 321.2 | −18.2 | 10.0 | 18.2 | 1.5 |
| 1.5 Kumuzhya dykes of second generation | $B_2$ | 300–500, >30 | 4/8/9 | 328.0 | 35.2 | 12.7 | 15.0* | 1.5 |
| **Voche–Lambina area, 67.7°N, 32.5°E** | | | | | | | | |
| 1.6 Ultramafic rocks, gneisses, amphibolites | A | 300–500, >30 | 3/9/13 | 270.9 | −23.5 | 11.8 | 13.8 | — |
| 1.7 " " " | A | 300–500, >30 | 4/9/10 | 80.4 | 31.2 | 12.7 | 14.0 | — |
| 1.8 " " " | A | 300–500, >30 | 7/18/23 | 265.8 | −27.5 | 12.0 | 9.5* | 1.0 |
| 1.9 " " " | C | 200–550 | 5/14/16 | 264.2 | 30.4 | 13.1 | 10.3* | 1.7 |
| 1.10 Peridotites, amphibolites, dolerites | $B_2$ | 200–550, >70 | 5/15/16 | 337.4 | 36.5 | 14.3 | 11.9 | 0.8 |
| 1.11 " " " | $B_2$ | 200–550 | 2/5/5 | 150.6 | −26.5 | 37.3 | 10.2 | 2.4 |
| 1.12 " " 1.10 + 1.11 | $B_2$ | 200–550, >70 | 7/20/21 | 335.5 | 33.9 | 16.4 | 7.7* | — |
| 1.13 Doleritic dykes | $B_1$ | 100–450, >20 | 2/8/11 | 322.5 | −27.3 | 14.1 | 13.2 | 1.4 |
| 1.14 " " " | $B_1$ | 100–450, >20 | 1/4/5 | 151.3 | 36.2 | 16.1 | 15.6 | 1.4 |
| 1.15 " " 1.13 + 1.14 | $B_1$ | 100–450, >20 | 3/12/16 | 325.7 | −30.9 | 14.9 | 10.1* | — |
| 1.16 " " Dyke 2 | A | 430–580 | 1/4/5 | 282.9 | −25.7 | 19.8 | 15.7 | 1.0 |
| **Kolvitsa area, 67.2°N, 32.6°E** | | | | | | | | |
| 1.17 Peridotites and porphyrites | A | 100–450, 0–11 | 2/4/4 | 249.7 | −36.3 | 10.7 | 21.4 | — |
| 1.18 " " " | A | 250–450 | 1/2/2 | 54.5 | 41.2 | — | — | — |
| 1.19 Peridotites and porphyrites, 1.17 + 1.18 | A | 100–450, 0–11 | 3/6/6 | 244.7 | −38.2 | 15.3 | 14.6* | 2.2 |
| 1.20 Peridorites and schists | $B_2$ | 450–530, 10–40 | 3/9/9 | 345.0 | 39.2 | 63.9 | 5.8* | 1.6 |
| 1.21 Porphyrites | $B_2$ | 450–530, 10–40 | 3/4/6 | 205.6 | −36.3 | 27.0 | 13.5 | — |

Notes: 1. $D^{(\circ)}$, $I^{(\circ)}$—mean declination and inclination; K—precision parameter, $\alpha_{95}^{(\circ)}$—95 per cent confidence circle radius for mean remanence directions, sample level statistics; S—ratio of K-value to that after tilt correction. 2.* used for pole calculations.

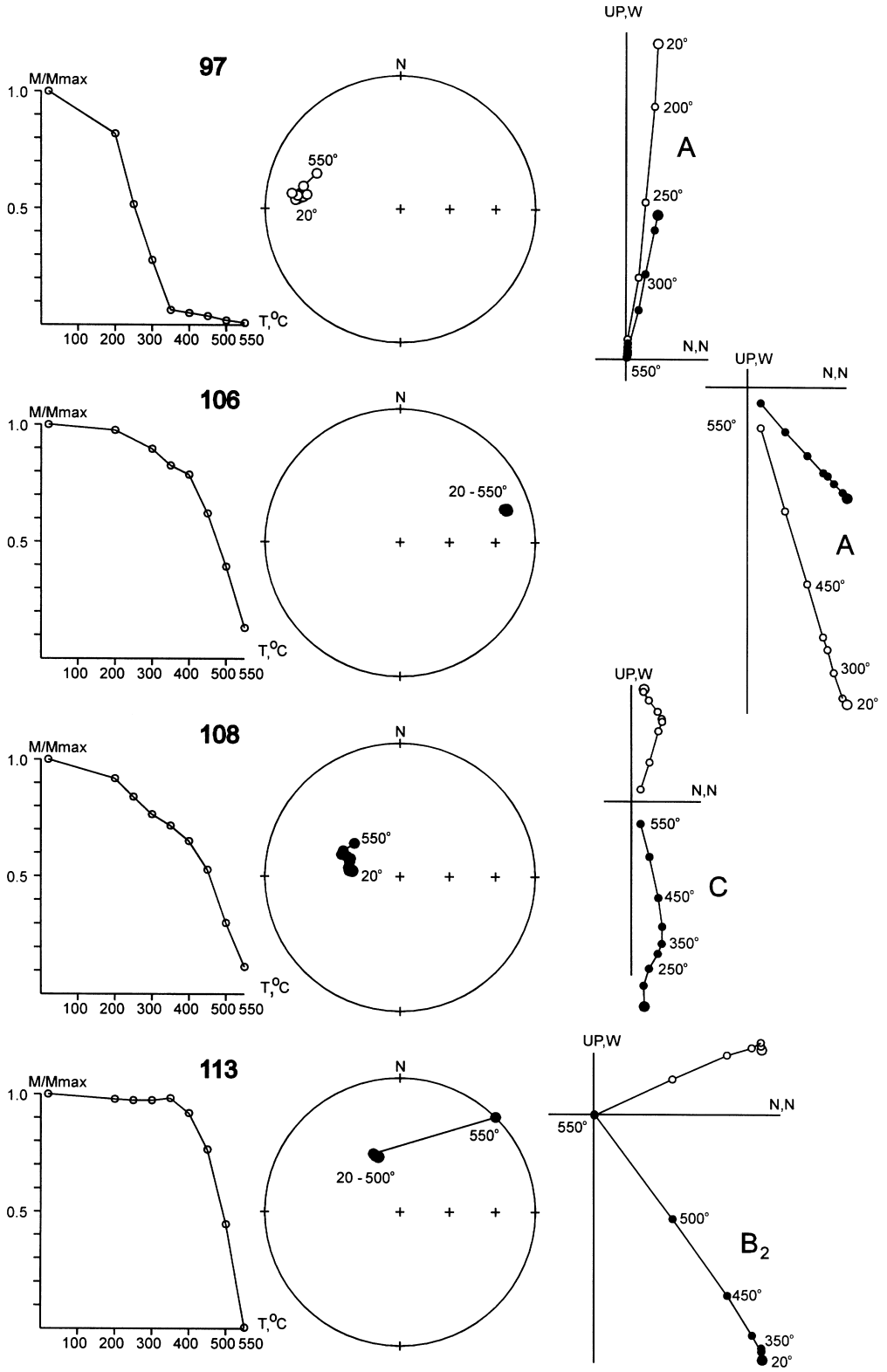



**Figure 5.** Demagnetization data for samples from the Moncha area. The plots illustrate the demagnetization characteristics in NRM-decay curves, equal area stereographic projections and orthogonal projections (Zijderveld diagrams). For stereographic projections, closed (open) symbols represent projections on the lower (upper) hemisphere. For the orthogonal projections, closed (open) symbols represent projections onto the vertical (horizontal) plane, where N is North, E is East, and UP is vertical up. Values are given in normalized units. NRM component indices (A, $B_1$, $B_2$, C) isolated during demagnetization are plotted behind the intervals of their isolation. Values of the temperatures (°C) and alternating field (mT) are also shown. Sample 97: Main Range gabbro, component A ($T_{ub} = 200$–$550$ °C), $D = 273.7°$, $I = -22.0°$. Sample 106: Monchegorsk pluton pyroxenite, component A ($T_{ub} = 200$–$550$ °C), $D = 72.9°$, $I = 17.7°$. Sample 108: Monchegorsk pluton picrite, component C ($T_{ub} = 400$–$550$ °C), $D = 287.3°$, $I = 54.3°$. Sample 113: Kumuzhya dyke dolerite, component $B_2$ ($T_{ub} = 200$–$550$ °C), $D = 336.6°$, $I = 52.7°$.

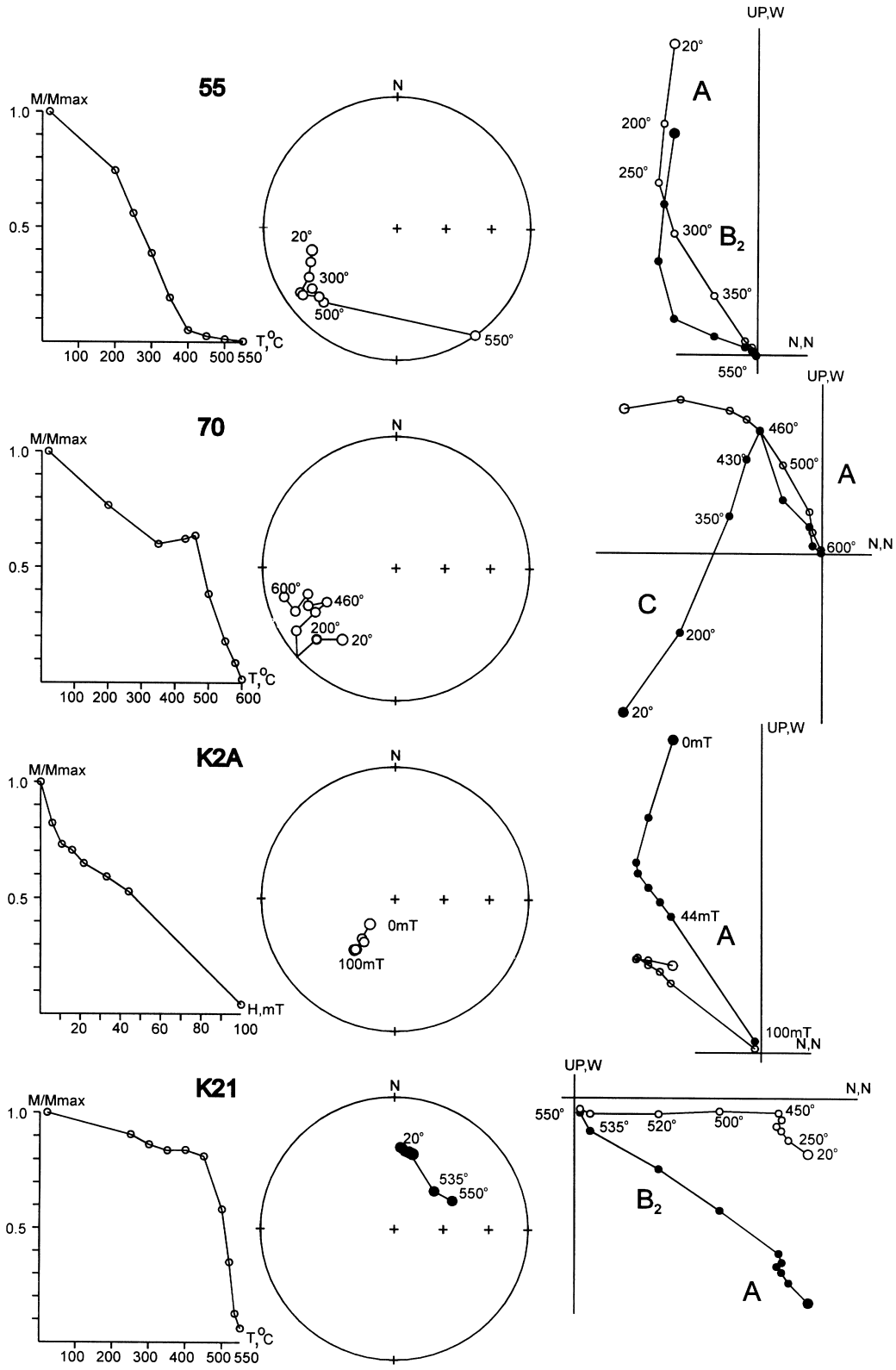



**Figure 6.** Demagnetization data for samples from the Voche–Lambina and Kolvitsa areas. Conventions and abbreviations are as for Fig. 5. Sample 55: amphibolite, component A ($T_{ub}$ = 20–300 °C), $D = 267.3°$, $I = -42.1°$ and component $B_2$ ($T_{ub}$ = 400–550 °C), $D = 227.9°$, $I = 12.1°$. Sample 70: dyke dolerite, component A ($T_{ub}$ = 500–580 °C), $D = 296.2°$, $I = -40.5°$. Sample K2A: porphyritic dyke, component $A_1$ ($H_b$ = 16–100 mT), $D = 209.1°$, $I = -49.0°$. Sample K21: porphyritic dyke, components A ($T_{ub}$ = 200–450 °C), $D = 73.8°$, $I = 47.0°$ and $B_2$ ($T_{ub}$ = 460–535 °C), $D = 0.0°$, $I = 33.0°$.

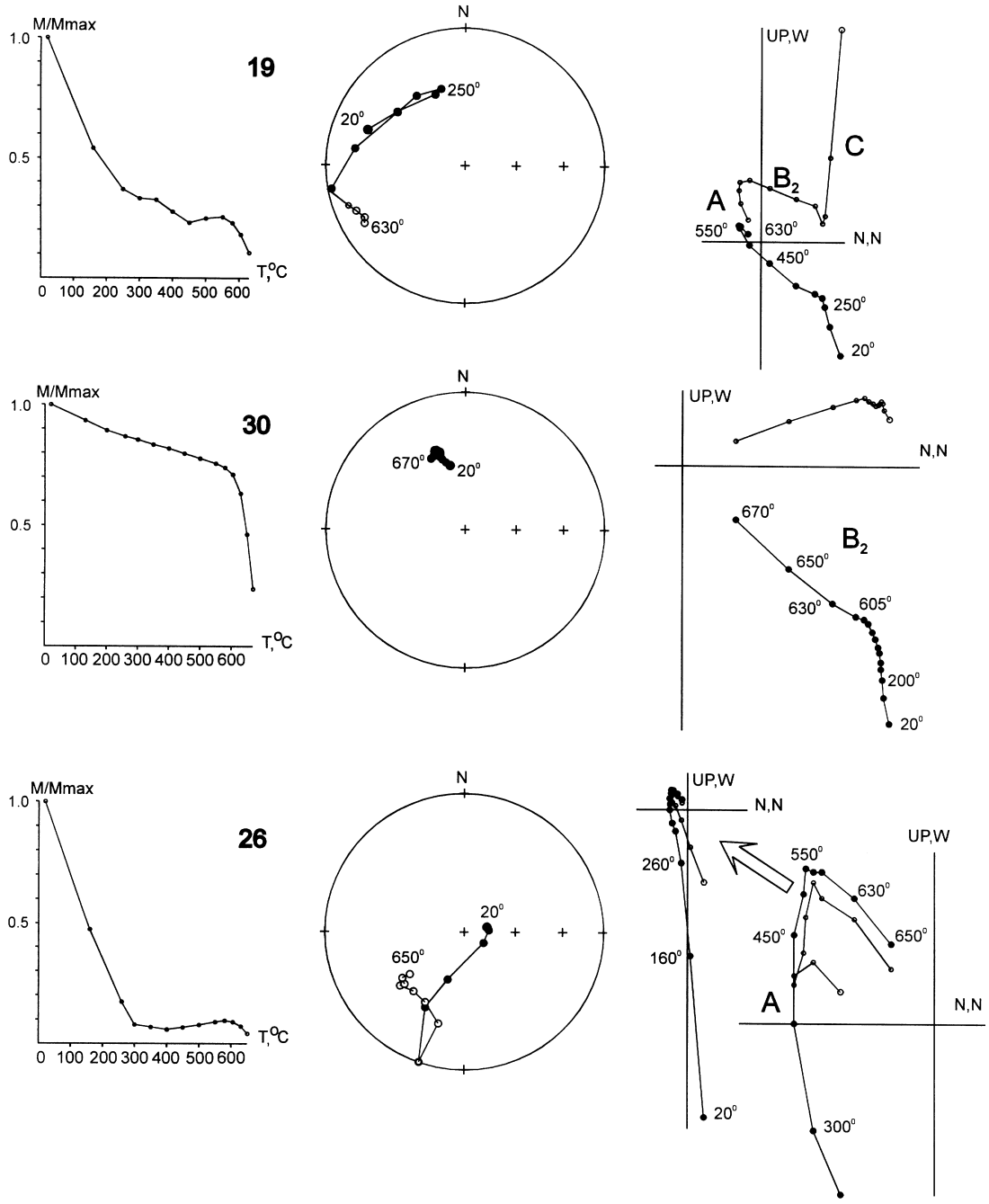



**Figure 7.** Demagnetization data for samples from the Matozero area. Conventions and abbreviations are as for Fig. 5. Sample 19: schist, components $B_2$ ($T_{ub} = 350$–$500\,^{\circ}C$), $D = 21.6^{\circ}$, $I = 35.5^{\circ}$ and $B_1$ ($T_{ub} = 580$–$630\,^{\circ}C$), $D = 241.3^{\circ}$, $I = -19.1^{\circ}$. Sample 30: gabbro, component $B_2$ ($T_{ub} = 630$–$670\,^{\circ}C$), $D = 339.9^{\circ}$, $I = 39.3^{\circ}$. Sample 26: dolerite, dual-polarity component A ($T_{ub} = 450$–$580\,^{\circ}C$), $D = 99.8^{\circ}$, $I = 33.1^{\circ}$ and ($T_{ub} = 580$–$650\,^{\circ}C$), $D = 227.2^{\circ}$, $I = -33.5^{\circ}$.

the A component in the Moncha and Voche–Lambina rocks (Fig. 10a). The second high-$T_{ub}$ dual-polarity component $B_2$ has a N-down/S-up direction (Table 1, Figs 6 and 10b).

The Matozero dolerites and schists carry four NRM components: the A-component ENE-down directions, isolated in the interval $T_{ub} = 300$–$550\,^{\circ}C$, two high-temperature components ($T_{ub} > 500\,^{\circ}C$), namely the $B_1$ (W-up) component and the $B_2$ (NW-down) component, and the C component (Fig. 7). The directions of these components calculated separately for dolerites and schists are very similar (Figs 11a and b). The intermediate-temperature and low-coercivity C component, with a W-down direction, has been isolated in specimens of all rock types except for the baked schists (Table 2, Fig. 11c).

Dolerites and gabbro, sampled in the Shoksha, Sheltozero and Ladva areas, represent different parts of the Ropruchey Sill. They have one dominant component of magnetization, with a maximum $T_{ub}$ spectrum at 450–580 °C and mean directions quite similar to those of the high-temperature $B_2$

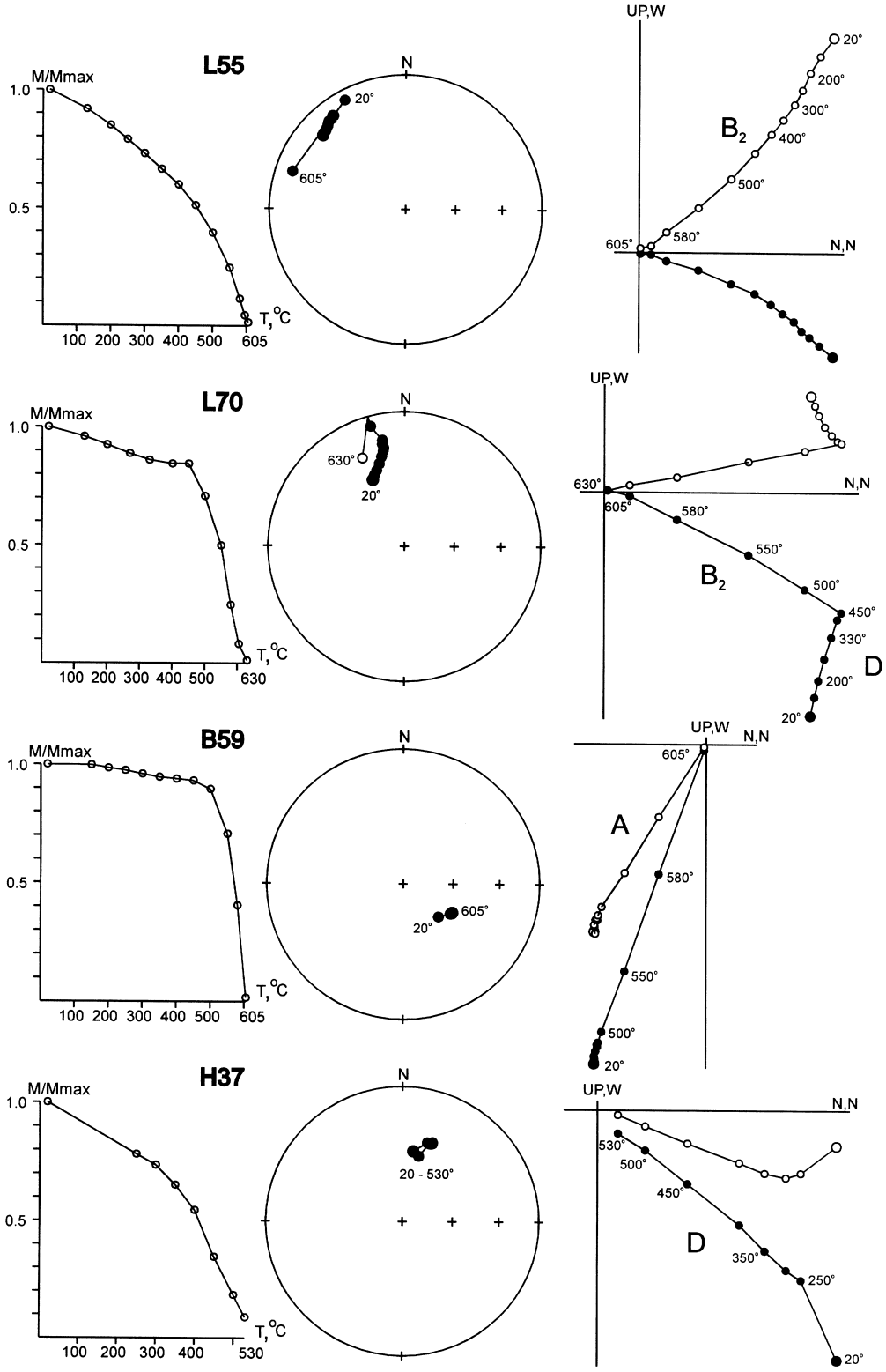



**Figure 8.** Demagnetization data for samples from the Ladva, Sheltozero, Burakovka and Hautavaara areas. Conventions and abbreviations are as for Fig. 5. Sample L55: dolerite, component $B_2$ ($T_{ub} = 550$–$650\,^{\circ}C$), $D = 323.1^{\circ}$, $I = 13.7^{\circ}$. Sample L70: gabbro, components $D$ ($T_{ub} = 20$–$450\,^{\circ}C$), $D = 237.4^{\circ}$, $I = 61.5^{\circ}$, and $B_2$ ($T_{ub} = 500$–$630\,^{\circ}C$), $D = 348.3^{\circ}$, $I = 25.0^{\circ}$. Sample B59: gabbro-norite, component A ($T_{ub} = 500$–$590\,^{\circ}C$), $D = 110.4^{\circ}$, $I = 27.2^{\circ}$. Sample H37: amphibolized gabbro, component D ($T_{ub} = 300$–$530\,^{\circ}C$), $D = 23.0^{\circ}$, $I = 27.0^{\circ}$.

component at Matozero. Sometimes it was possible to isolate an intermediate-temperature component D (SW-down). This is the only component for the monzonite veins, which cut the sill in the Sheltozero area (Table 2, Figs 8, 12a, b and c).

Two components isolated in the Burakovka layered intrusion are associated with different blocks: component $A_1$ with the Aganozero block, and component $A_2$ with the Burakovka and Central blocks. All these components are single polarity, with

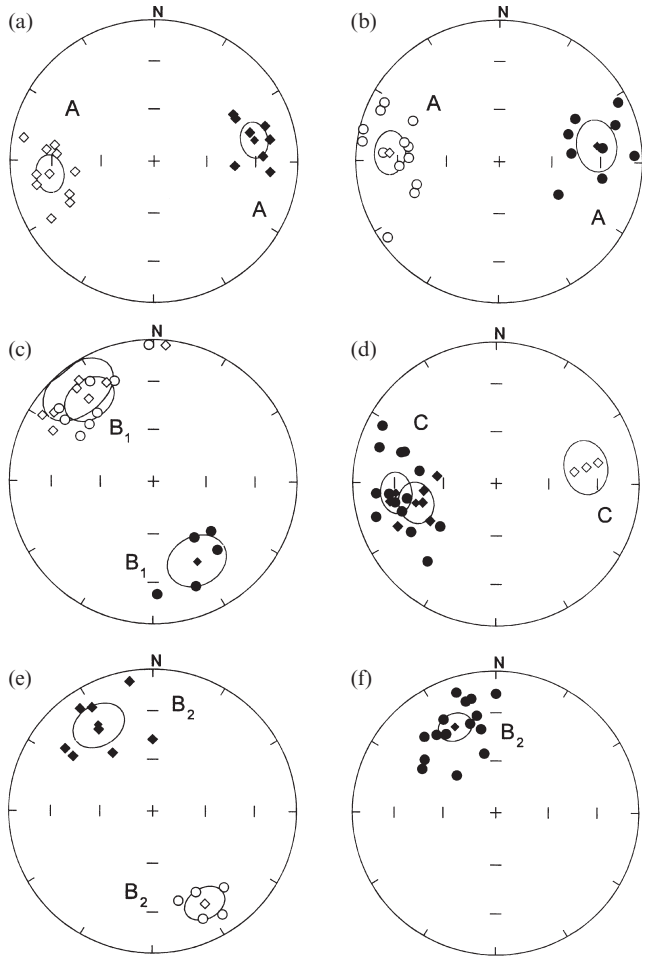



**Figure 9.** Characteristic magnetization directions (ChRM) for the Moncha (diamonds) and Voche–Lambina (circles) rocks. Closed and open figures represent projections onto the lower and upper hemispheres, respectively. Diamonds with ovals represent mean direction and circles of 95 per cent confidence. Equal-area projections. (a) Monchegorsk pluton pyroxenites and gabbro-dolerites of the Main Range, component A; (b) Voche–Lambina ultramafites and metamorphic country rocks and dyke 2, component A; (c) Voche–Lambina and Moncha dolerite dykes, component $B_1$; (d) Voche–Lambina and Moncha picrites and country rocks, component C; (e, f) Kumuzhya dykes of second generation and altered Voche–Lambina rocks, component $B_2$.

similar inclinations ($I = 54.0^\circ$ to $65.4^\circ$) but with declinations varying from $D = 69.5^\circ$ for the Aganozero block to $D = 135.7^\circ$ for the Central block (Figs 8, 12d, e and f, Table 2).

Hautavaara amphibolized gabbro and andesites carry three NRM components (Figs 10c and d): the A component with a NE-down direction, the $B_2$ component with a NW-down direction, and the D component with a NNE-down direction. All of these components are isolated in the range 10–50 mT; however, they have different ranges for $T_{ub}$ (Table 2).

## 4.2 Magnetic carriers and their origin

This part of paper presents the results of thermomagnetic analyses carried out on more than 100 samples, in order to identify their magnetic carriers, their origin and possible time of creation.

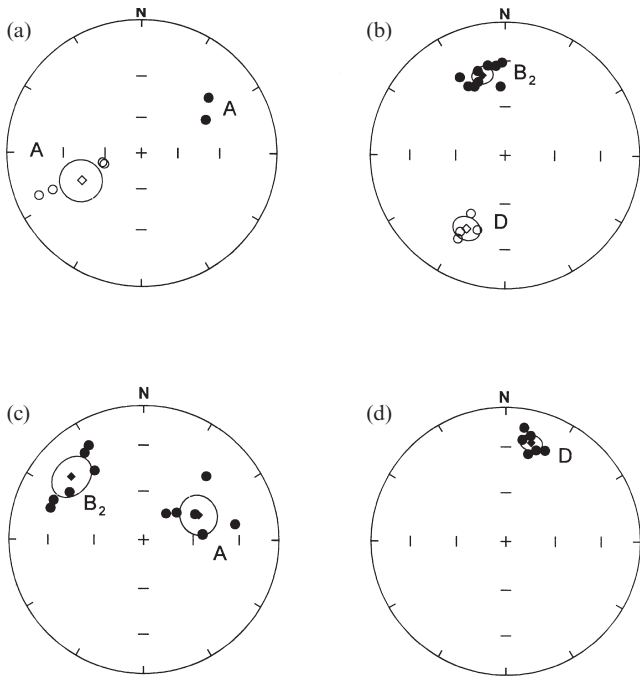



**Figure 10.** Directions of ChRM for Kolvitsa (a, b) and Hautavaara (c, d) amphibolites, gabbro and porphyrites. (a) component A; (b) components $B_2$ and D; (c) components A and $B_2$; (d) component D. Other conventions as in Fig. 9.

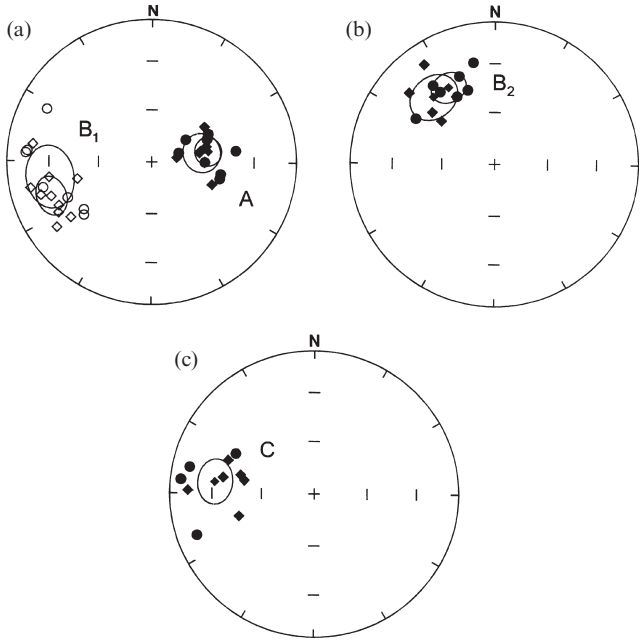



**Figure 11.** Directions of ChRM for Northern Onega Jatulian dolerites (circles), and schists (diamonds). (a) Matozero dolerites and schists, components $B_1$ and A; (b) Matozero dolerites and schists, component $B_2$; (c) Matozero dolerites and schists, component C.

### *4.2.1 Moncha area*

Thermomagnetic analysis of pyroxenite and peridotite samples revealed the presence of virtually pure magnetite only (Fig. 13). Normally, magnetite occurs at the late magmatic stage, at temperatures above its Curie point (580 °C), and is thereby

**Table 2.** Mean remanence directions for Karelian rocks.

| Numbers Rock unit | NRM component | Isolation interval °C, mT | Sites/ /samples/ /specimens | $D^{(\circ)}$ | $I^{(\circ)}$ | K | $\alpha_{95}^{(\circ)}$ |
|---|---|---|---|---|---|---|---|
| **Matozero area, 62.5°N, 35.5°E** | | | | | | | |
| 2.1 Dolerites | $B_2$ | >500, 40–90 | 2/7/11 | 329.1 | 37.4 | 32.5 | 9.3* |
| 2.2 Schists | $B_2$ | >580 | 2/5/5 | 317.9 | 36.7 | 22.6 | 13.2 |
| 2.3 Doleritic dykes | $B_1$ | >580 | 2/8/12 | 257.6 | −31.1 | 11.4 | 14.7* |
| 2.4 Baked schists | $B_1$ | >580 | 2/8/11 | 249.8 | −27.3 | 25.7 | 9.8* |
| 2.5 Doleritic sill | A | 300–550, >60 | 3/9/11 | 78.9 | 57.3 | 34.8 | 7.9* |
| 2.6 Host schists | A | 300–550 | 2/5/7 | 76.6 | 60.9 | 32.7 | 11.0* |
| 2.7 Dolerites and gabbro | C | 250–400, 20–90 | 2/4/6 | 275.4 | 19.4 | 12.9 | 19.5 |
| 2.8 Schists | C | 250–580, 5–40 | 3/6/10 | 276.8 | 40.3 | 25.0 | 11.4 |
| 2.9 Matozero rocks, 2.7 + 2.8 | C | 250–580, 5–90 | 5/10/16 | 276.1 | 32.1 | 15.4 | 11.3* |
| **Ropruchey Sill areas: Shoksha and Sheltozero, 61.2°N, 35,6°E; Ladva, 61.5°N, 34.5°E** | | | | | | | |
| 2.10 Shoksha and Sheltozero gabbro-dolerites | $B_2$ | 450–580 | 2/6/9 | 349.0 | 23.7 | 30.0 | 10.4* |
| 2.11 Shoksha and Sheltozero dolerites | $B_2$ | 450–580 | 3/6/10 | 5.7 | 14.9 | 52.5 | 7.9* |
| 2.12 Ladva gabbro-dolerites | $B_2$ | 450–580 | 4/11/17 | 357.1 | 23.4 | 35.6 | 7.1* |
| 2.13 Sheltozero and Ladva gabbro-dolerites | D | 200–350 | 4/9/9 | 253.1 | 70.4 | 35.6 | 7.8* |
| 2.14 Sheltozero monzonites | D | 100–250 | 4/4/7 | 212.2 | 10.8 | 34.9 | 11.9 |
| **Burakovka area, 62.1°N, 36.2°E** | | | | | | | |
| 2.15 Central block, gabbro-norites | A | 450–590 | 5/16/18 | 119.6 | 63.5 | 35.9 | 5.8* |
| 2.16 Burakovka block, gabbro-diorites | A | 450–550, 20–70 | 4/9/9 | 135.7 | 65.4 | 44.2 | 7.0* |
| 2.17 Aganozero block, ultramafic rocks | A | 450–580 | 4/12/16 | 69.5 | 54.0 | 38.7 | 6.5* |
| **Hautavaara area, 61.8°N, 32.9°E** | | | | | | | |
| 2.18 Amphibolized gabbro and andesites | A | 250–450, >20 | 6/7/7 | 65.4 | 53.1 | 20.1 | 11.8* |
| 2.19 " " " | $B_2$ | >500, 10–45 | 4/6/8 | 311.4 | 29.5 | 22.4 | 12.1* |
| 2.20 " " " | D | 300–530, 35–55 | 5/6/6 | 14.2 | 24.8 | 99.3 | 5.7* |

Notes: see notes for Table 1.

capable of carrying a thermoremanent magnetization synchronous with the rock formation; however, magnetite more often contains a palaeomagnetic record of later processes (e.g. Pisakin *et al.* 1992). A possible magmatic origin of the Monchegorsk pluton magnetite is indicated by its stability during repeated heating–cooling cycles (a characteristic typical of secondary magnetite, such as that produced by serpentinization), and by the similarity of the $J_{rs}(T)$ and $J_n(T)$ dependences (compare Fig. 5, sample 106 and Fig. 13, sample 111). Therefore, we attribute a primary origin to the A component in the ultramafics of the Monchegorsk pluton.

### 4.2.2 *Voche–Lambina area*

Thermomagnetic analysis of dolerites gave the most important results. The main magnetic carrier is magnetite, both for $J_n$ and $J_{rs}$ (Fig. 6, sample 70 and Fig. 13, sample 35). However, a comparison of the dependences of $J_s(T)$ and $J_{rs}(T)$ from the first and second heating cycles shows partial homogenization of exsolved titanomagnetite accompanied by a decrease of the Curie point ($T_c$). These dependences suggest that the magnetite matrix includes relicts of titanomagnetite with a $T_c$ of about 250–350 °C. Microprobe results support the conclusion that the $T_c$ of primary magnetite (10–20 per cent $TiO_2$) was nearly the same. Therefore, the NRM component $B_1$, which is recognized from the thermal demagnetization in approximately the same temperature interval, may be synchronous with the formation of dolerites. The thermomagnetic analysis of peridotites also shows that they may contain rock-syngenetic relicts of a magnetite group mineral (1–1.5 per cent Mg, 3.7–5.3 per cent Cr and 1–1.5 per cent Al according to microprobe analysis).

### 4.2.3 *Kolvitsa area*

The magnetic carrier in the Kolvitsa peridotites is virtually pure magnetite (Fig. 13, sample K17). The shape of the thermomagnetic curves $J_{rs}(T)$ is controlled by the spectra of the unblocking temperatures $T_{ub}$. Two magnetic components are recognized from the $\Delta J_{rs}(T)$ and $J_n(T)$ dependences; however, the maximum $T_{ub}$ spectra for SIRM is shifted to lower temperatures with respect to that for NRM (Figs 6 and 13 for sample K21). Thermal instability of the low-temperature carrier argues against its primary magmatic origin. In contrast, the porphyrites (sample K21) are likely to contain a primary magnetic mineralogy.

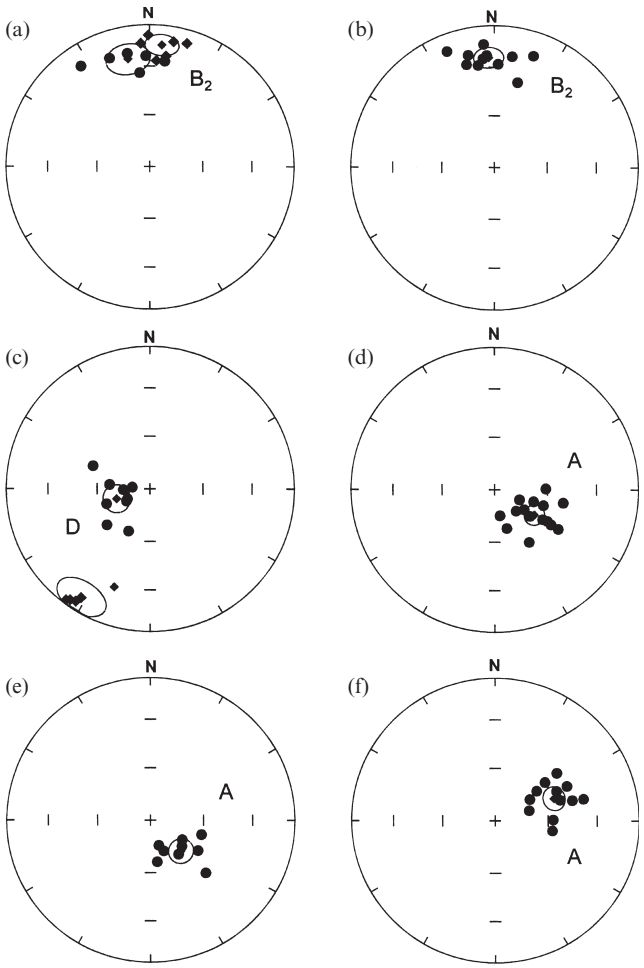



**Figure 12.** Directions of ChRM for Southern Onega rocks. (a) Shoksha and Sheltozero gabbro-dolerites (circles) and dolerites (diamonds), component $B_2$; (b) Ladva gabbro-dolerites, component $B_2$; (c) Sheltozero and Ladva gabbro-dolerites (circles) and monzonites (diamonds), component D; (d) Burakovka layered intrusion, Central block, gabbro-norites; (e) Burakovka, Burakovka block, gabbro-diorites; (f) Burakovka, Aganozero block, ultramafics.

#### *4.2.4 Matozero area*

The values of magnetic hardness $Q_{rs}$ in the strongly magnetic gabbro and dolerites do not exceed 160, indicating the presence of magnetically soft magnetite. Such magnetite is the product of rather intense, metamorphic and metasomatic alterations, resulting in complete recrystallization of the primary magnetite and the formation of magnetically soft, large-grained, virtually pure secondary magnetite. Therefore, our palaeomagnetic data from these strongly magnetic rocks may be related to a metamorphic event, presumably of Svecofennian age. In contrast, the weakly magnetic lithologies contain fine-grained magnetite of very high magnetic hardness ($Q_{rs} = 3H_c > 1500$) with nearly single-domain-sized grains, and are capable of preserving their primary NRM for a long time. These dolerites carry the A component in magnetite.

#### *4.2.5 Ladva, Shoksha and Sheltozero areas*

Outcrops in the Ladva area are represented by samples exhibiting intense homogenization upon heating in air (Fig. 13, sample M-1), which suggests insignificant alterations and the presence of relicts of a primary magnetic mineral (high-Ti titanomagnetite). The thermomagnetic analysis of dolerites from the Shoksha and Sheltozero areas gave similar, but not so strongly convincing, results (Fig. 13, sample B91).

#### *4.2.6 Burakovka area*

Ultramafites are found amongst the samples from the Aganozero block of the Burakovka layered intrusion. Thermomagnetic analysis of these samples gives results that are characteristic for magnesium magnetite with a small percentage of MgO, which indicates a primary origin for this magnetite. On the other hand, petrographic and microprobe investigations have established the presence of secondary olivine-generated magnetite, which appeared during serpentinization of the Aganozero peridotites (Pisakin *et al.* 1992). Thermomagnetic analysis of gabbro-norites yields $J_{rs}(T)$ dependences characteristic of virtually pure magnetite, whereas the decay curves $J_n(T)$ (see Fig. 8, sample B59), in conjunction with microscopic and microprobe analyses, indicate a thermoremanent origin of the NRM in the gabbro-norites. These latter rocks, which contain so-called 'cloudy' plagioclase, yield results that are consistent with the formation of magnetite within such plagioclase during the consolidation of the Burakovka intrusion (Pisakin *et al.* 1992).

To summarize, the thermomagnetic and (less extensive) microprobe analyses indicate that many of the studied mafic and ultramafic rocks of the Karelia–Kola region contain a ferrimagnetic mineral of the magnetite group, syngenetic with and synchronous with the formation of these rocks: magnesium-aluminium-chrome magnetite in most ultramafic and mafic rocks, and titanomagnetite, with appreciable amounts of magnesium and aluminium, in dolerites. Such a mineralogy appears most often in the stage of exsolution, but the thermomagnetic data suggest the presence of relicts of the primary magnetic mineralogy that record the formation conditions and palaeomagnetic setting of these rocks.

### 4.3 Dating of the NRM components

Tilt corrections (unfolding), based on the structural attitudes of the layers near the boundaries of the lopolith-like bodies, as well as on the foliation and banding of country metamorphic rocks, lead to a decrease in the values of the precision parameter $K$ by factor of 1.5–6.5 (Table 1). One exception is the data set from Voche–Lambina, where $S = K/K_{corr} \approx 1$ due to the monoclinal structure of the area (Fig. 2). High values of the ratio $S$ characterize the between-group statistics too: $S = 6.8$ for the A component (entries 1.2, 1.3, 1.8, 1.16 and 1.19 in the Tables); $S = 10.9$ for the $B_1$ component (1.4, 1.13 and 1.14 in the Tables) and $S = 4.5$ for the $B_2$ component (1.5, 1.10, 1.11 and 1.20 in the Tables). The negative results of the tilt corrections mean that the NRM components of the intrusive and metamorphic rocks investigated here are younger than the structural pattern created during the first, most intensive rock deformation. However, they do not exclude a primary, thermoremanent origin for the NRM components of the intrusive bodies.

The primary origin of some of the NRM components, as evidenced by the thermomagnetic and microprobe data, enables us to estimate the ages of these components directly from the ages of the corresponding rocks (if these rocks have

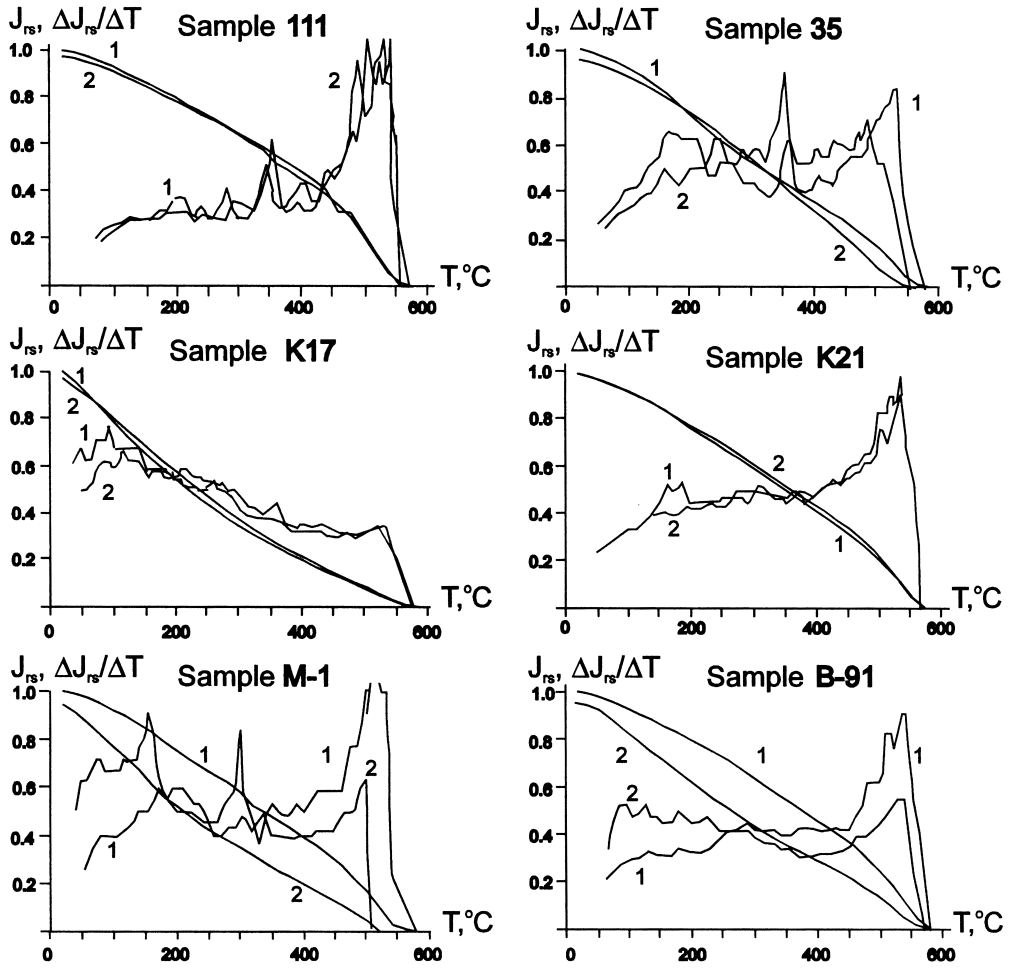



**Figure 13.** Thermomagnetic and differential thermomagnetic curves: decay of the SIRM, per cent per 25 °C. Sample 111: Monchegorsk pluton pyroxenite; sample 35: Voche–Lambina dolerite; sample K17: Kolvitsa peridotite; sample K21: Kolvitsa porphyritic dyke; sample M1: Ladva gabbro-dolerite; sample B91: Shoksha dolerite. 1: first heating, 2: second heating.

been radiometrically dated). For the Kola and Belomorian provinces, the A component in the Monchegorsk pluton pyroxenites is dated from a U–Pb age of $2493 \pm 7$ Ma (Table 1, entry 1.2), whereas in the Kolvitsa porphyrities it is dated from a U–Pb age of $2450 \pm 10$ Ma. Two metamorphic events, which yield Rb–Sr ages of $2450 \pm 83$ Ma and $2198 \pm 172$ Ma, are possibly related to the A and C components isolated in the Voche–Lambina rocks (Table 1, entries 1.6, 1.7, 1.8 and 1.9). In the Karelian province, the $B_1$ component from the Matozero doleritic dykes is dated from a U–Pb age of $2150 \pm 60$ Ma (Table 2, entries 2.3 and 2.4), and the $B_2$ component from the Ropruchey Sill is dated from a U–Pb age of $1770 \pm 12$ Ma (Table 2, entries 2.10, 2.11 and 2.12). The A component from the Burakovka gabbro-norites is dated from a U–Pb age of $2449 \pm 1$ Ma (Table 2, entries 2.15 and 2.16), a similar age to that estimated in the Kola and Belmorian provinces above.

The other NRM components that we isolated have been dated indirectly; that is, using structural and stratigraphic relationships which clarify the sequence of origin and timing of those components.

The A component isolated in the Moncha area is in the oldest rocks (i.e. the pyroxenite intrusion and country gabbro-dolerite that emplaces the pluton). The U–Pb isochron age of the gabbro-dolerite is $2436 \pm 25$ Ma. A reversal test for the A component isolated in the Moncha rocks gave a positive result (class C), with an antipodal angle of (180–5)° between the mean directions for the normal and reverse magnetizations (Table 1, entries 1.2 and 1.3). This result implies a short time span between these magnetizations. Hence, the age of the A component for the Main Range rocks is nearly the same as the age of pluton pyroxenites. The structural relationships between these bodies indicate that the Monchegorsk pluton cannot be older than the Main Range gabbro-dolerites. Therefore, we estimate the age of the A component, isolated in the pluton pyroxenites, as $2406 \pm 3$ Ma—the age of the Main Range amphibolized gabbro (Mitrofanov *et al.* 1993). Field relationships between the Moncha picritic dykes (entry 1.1) and the doleritic dykes of first and second generations (entries 1.4 and 1.5) indicate the following time succession of NRM component acquisition: A, C, $B_1$, $B_2$. Taking into consideration that the possible ages of the dykes carrying $B_1$ and $B_2$ components are 2200 and 1800 Ma, it is possible to estimate the age of the C component as 2200–2250 Ma, and the younger age limit of the A component as 2250 Ma. In the Voche–Lambina area, the A component was isolated only in the country metamorphic rocks and in the oldest Dyke 2, whereas the C component is associated with zones of younger metamorphism. The $B_1$ component is recorded only in the doleritic

dykes cutting these metamorphic rocks. Finally, the $B_2$ component has been isolated in all rock types mentioned above and, thus, appears to be superimposed. Hence, in this case we have the same time succession of NRM-component acquisition as in the Moncha area; that is: A, C, $B_1$, $B_2$. The reversal test for the A component in the Voche–Lambina rocks (entries 1.6–1.8) was found to be indeterminate (class I), with an angle of (180–12)° between the mean directions for the normal and reverse magnetizations. This result, together with the considerable differences between the mean directions of the A component for the Moncha, Voche–Lambina and Kolvitsa areas, may possibly be related to the variety of ages of this component in the mentioned areas. These younger age limits of components A and C are indirectly confirmed by the fact that these components were recorded in the Lower Jatulian formations of the North Onega region (Table 2) and are absent in the younger rocks (i.e. this limit is 2250–2300 Ma too).

In the North Onega region, component $B_1$ was isolated in the Upper Jatulian doleritic dykes with an age of 2100–2200 Ma, and in the Lower Jatulian baked quartzites and schists; in the latter units this component represents a high-temperature haematite component (Table 2, entries 2.3 and 2.4). The Lower Jatulian schists and subvolcanic dolerites that are situated far from the doleritic dykes of the third phase carry a magnetite component A (Table 2, entries 2.5 and 2.6) and an intermediate-temperature component C (entries 2.7, 2.8 and 2.9). Thus, it appears that the $B_1$ component is a primary one according to the positive baked contact test and is of Late Jatulian age (i.e. 2100–2200 Ma).

Component $B_2$ is the single high-temperature component ($T_{ub} = 450$–$580\,^{\circ}C$) isolated in the Ropruchey dolerites and gabbro (Table 2, entries 2.10, 2.11 and 2.12). This fact, together with the absence of metamorphic features, prompts us to conclude a probable primary origin of the $B_2$ component in these rocks. Hence, its age is approximately 1800 Ma, based on a U–Pb age of $1770 \pm 12$ Ma for these rocks, whereas in older rocks the $B_2$ component appears to be a secondary superimposed component of NRM. Component D, detected in some samples from the Sheltozero area (Table 2, entries 2.13 and 2.14) is thought to be a younger secondary component too. However, its age cannot be estimated as yet.

We consider the NRM components of the Hautavaara area Archaean rocks to be secondary ones, acquired during the Sariolian (2400 Ma; component A) and Svecofennian (1880 Ma; component $B_2$) metamorphic events. We cannot exclude the possibility of a thermal influence from the intrusion of the neighbouring Ulalega Rapakivi Massif ($1543 \pm 8$ Ma; Neymark *et al.* 1994) having reset the magnetization D (Table 2, entries 2.18–2.20).

## 5 ASSOCIATED NEW RESULTS

In addition to the results presented above, some new palaeomagnetic data have been obtained through the investigation of other intrusive bodies and host rocks of the Belomorian and Karelian provinces. These data have been published recently (Khramov *et al.* 1997)—they are updated here but are presented separately because of the uncertainty in their ages or poor statistics.

The Tolstik area features a gabbroic–dioritic layered intrusion of $2434 \pm 7$ Ma, succeeded by a $2405 \pm 20$ Ma intrusion of potassic granites (Bogdanova & Bibikova 1993). This lopolith-like body is 6 km long and up to 2 km wide and is mantled by a broad rim of banded garnet amphibolites. In addition, there are three generations of mafic dykes; the youngest of them is of Svecofennian age (Miller *et al.* 1995). Samples were taken from the layered intrusion, its rim amphibolites and from the youngest gabbroic dykes. A dual-polarity A component, with intermediate $T_{ub}$ values of 200–450 °C is isolated in the gabbro from the layered intrusion and its amphibolites. A high-temperature $B_2$ component is typical for both dykes and amphibolites of the contact. The mean directions of these components are listed in Table 3.

The Segozero area (Fig. 3) extends along the southern shore of Lake Segozero, close to the northern margin of the Onega depression. Doleritic dykes here carry two NRM components: the first one, indicated as $B_1$, is isolated in the range 300–580 °C; the second component $B_2$ is a high-temperature and high-coercivity one. The doleritic sill carries only one intermediate-temperature, low-coercivity component A (Table 5).

The Girvas area is about 70 km to the south of Segozero, near the Girvas waterfall (Fig. 3). The doleritic dykes here are characterized by an A component, with intermediate values of $T_{ub}$, and a high-temperature component $B_2$. The basaltic flows and schists carry the intermediate-temperature component D and three high-temperature components $B_1$, $B_2$ and C (Table 3).

There are no direct radiometric ages for these Karelian rocks. However, the stratigraphic positions of sampled bodies and the $T_{ub}$ spectra of their NRM components are the same as for the Matozero rocks. Hence, approximate ages of the isolated component can be estimated. These are: A ~ 2250 Ma; C ~ 2200 Ma; $B_1$ ~ 2150 Ma; $B_2$ ~ 1800 Ma. We attribute the A component of the Tolstik rocks to the most intense metamorphic event, terminated by the intrusion of potassic granites $2405 \pm 20$ Myr ago (Bogdanova 1996).

## 6 DISCUSSION OF THE RESULTS AND CONCLUSIONS

This study identifies 36 independent and statistically valid ($N > 6$, $\alpha_{95} < 15°$) ChRM directions, which were grouped in five sets according to their affinity to the rock types, blocking temperatures and coercive spectra. These sets were named as the A, $B_1$, $B_2$, C and D components and their ages have been estimated. The presence of many of them within each area, often within the same geological body and, in some cases, within the same samples reveals a complicated magnetization history, caused by intense magmatic and metamorphic events. In the Kola and Belomorian provinces there was mafic-ultramafic magmatism during the period 2400–2250 Ma, accompanied by a common regional metamorphism: two stages of this process are implied in the magnetization components A and C. These events were followed by Late Jatulian (component $B_1$) and Svecofennian (component $B_2$) events of crustal activation. The same events are also recorded magnetically in the Karelian province; however, records of the Jatulian (components C and $B_1$) and especially Svecofennian (component $B_2$) events dominate there. Moreover, in some cases (entries 2.13, 2.14, 2.20 and 3.10) younger remagnetization events have also been recorded. These are probably related to post-Svecofennian and Jotnian activity. This sequence of

**Table 3.** Mean remanence directions of rocks from the Belomorian and Karelian.provinces: associated results.

| Numbers Rock unit | NRM component | Isolation interval °C, mT | Sites/ /samples/ /specimens | D(°) | I(°) | K | $\alpha_{95}^{(\circ)}$ |
|---|---|---|---|---|---|---|---|
| **Tolstik area, 66.8°N, 32.4°E** | | | | | | | |
| 3.1 Layered intrusion and amphibolites | A | 200–450 | 4/5/6 | 252.1 | −42.7 | 12.6 | 12.6 |
| 3.2 " " " | A | 200–450, >40 | 3/5/5 | 77.1 | 50.5 | 13.7 | 16.9** |
| 3.3 " " " 3.1 + 3.2 | A | 200–450, >40 | 7/10/11 | 254.4 | −46.7 | 14.1 | 11.8* |
| 3.4 Gabbroic dykes and amphibolites | $B_2$ | 450–600 | 3/7/8 | 337.1 | 37.6 | 25.8 | 10.4* |
| **Segozero area, 63.2°N, 33.5°E** | | | | | | | |
| 3.5 Doleritic dykes | $B_2$ | >605, >100 | 2/7/11 | 325.4 | 44.6 | 14.1 | 14.1* |
| 3.6 " " " | $B_1$ | 300–580 | 2/5/5 | 290.1 | −25.6 | 16.4 | 15.5* |
| 3.7 " " sill | A | 300–580, 20–100 | 2/7/10 | 62.6 | 52.5 | 20.5 | 11.6* |
| **Girvas area, 62.5°N, 33.5°E** | | | | | | | |
| 3.8 Dolerite dykes | $B_2$ | 500–630 | 1/3/4 | 1.3 | 20.1 | 26.1 | 15.8 |
| 3.9 " " " | A | 300–500, >60 | 3/8/11 | 94.4 | 57.0 | 19.8 | 11.1* |
| 3.10 Flows and schists | D | 480–580 | 2/6/7 | 256.1 | 67.0 | 65.2 | 7.1 |
| 3.11 " " " | C | >580 | 1/4/4 | 289.7 | 10.8 | 14.2 | 18.6 |
| 3.12 " " " | $B_1$ | >580 | 1/3/3 | 161.6 | 10.2 | 11.4 | 23.9 |
| 3.13 " " " | $B_2$ | >580 | 2/6/10 | 321.3 | 39.5 | 36.3 | 9.5* |

Notes: 1. ** These data include results from (Krasnova & Gooskova 1997). 2. Other notes as for Table 1.

magnetization events generally does not contradict the well-known tectonomagmatic history of the eastern Fennoscandian Shield (Gaal & Gorbatschev 1987; Gorbatschev & Bogdanova 1993). There is one exception, however: the supposed post-Svecofennian and Jotnian reactivations of the Southern Karelian crust. Manifestations of these events in the geological record have not been recognized in this part of the shield as yet, although palaeomagnetic records of such late events have been recently revealed (Damm *et al.* 1997).

Palaeomagnetic poles, calculated from ChRM directions obtained in this study (Tables 4 and 5) form roughly two bands: one along 45°N and a submeridional one along 310°E (Fig. 14). The first band, obtained recently for the Karelian Province only (Damm *et al.* 1997), appears to be a good fit to the Svecofennian track of the APWP drawn by Elming *et al.* (1993) on the basis of the revised and updated palaeomagnetic database for Fennoscandia (Elming *et al.* 1996), although, for the above-mentioned time interval, it does not differ significantly from that proposed earlier (Pesonen & Neuvonen 1981). However, the new pre-Svecofennian poles of this study are evidently shifted eastwards from the APWP drawn by Elming *et al.*, with the exception of poles 2.15, 2.16 and 3.9, which are located near that APWP. Note also that two poles, 1.4 and 1.15, are located in the lower northern latitudes to the west of the APWP. They have an age of 2150 Ma, possibly indicating a 'cusp' in the APWP. It might also be possible that this deviation represents local tectonic rotations in the Moncha and Voche–Lambina areas.

When interpreting the palaeomagnetic results from the Burakovka layered intrusion the local tectonic setting must

**Table 4.** Palaeomagnetic poles obtained for the Kola and Belomorian provinces, eastern Fennoscandian Shield—this work.

| Numbers | Rock unit Λ, °E | NRM component A | Radiometric age, Ma $_{95}^{\circ}$ | Assigned magnetic age, Ma | Polarity | Pole position Φ | ,°N | |
|---|---|---|---|---|---|---|---|---|
| 1.1 | Monchegorsk pluton picrites | C | | 2250–2150 | 8N | 16.6 | 323.3 | 8.1 |
| 1.2 | Monchegorsk pluton pyroxenites | A | 2493 ± 7 | 2490–2406 | 8R | −19.4 | 309.0 | 9.6* |
| 1.3 | Main Range gabbro-dolerites | A | 2436 ± 25 | 2460–2410 | 10N | −16.7 | 304.8 | 8.3* |
| 1.4 | Monchegorsk dolerite dyke | $B_1$ | | 2150–2000 | 6N | 7.9 | 251.1 | 14.6 |
| 1.5 | Kumuzhya dykes | $B_2$ | | 2000–1800 | 8N | 37.4 | 251.5 | 13.2 |
| 1.8 | Voche–Lambina rocks | A | 2450 ± 83 | 2400–2200 | 9N, 9R | −15.1 | 300.7 | 7.7* |
| 1.9 | Voche–Lambina rocks | C | 2198 ± 172 | 2300–2100 | 14N | 12.9 | 314.1 | 8.6* |
| 1.12 | Voche–Lambina rocks | $B_2$ | | 1900–1800 | 15N, 5R | 38.5 | 242.6 | 6.6 |
| 1.15 | Voche–Lambina dolerite dykes | $B_1$ | | 2150–2000 | 8N, 2R | 2.0 | 245.2 | 8.4 |
| 1.19 | Kolvitsa rocks | A | 2423 ± 3 | 2426–2414 | 4N, 2R | −29.0 | 317.4 | 13.3* |
| 1.20 | Kolvitsa rocks | $B_2$ | | 1900–1800 | 9N | 44.2 | 232.6 | 5.4 |
| 3.3 | Tolstik gabbro intrusion | A | 2434 ± 7 | 2440–2405 | 5N, 5R | −31.6 | 304.7 | 12.2 |
| 3.4 | Tolstik dykes and amphibolities | $B_2$ | | 1950–1800 | 7N | 42.0 | 241.6 | 9.4 |

Notes: 1. Pole numbers are the same as in Tables 1, 2 and 3. Φ, Λ: latitudes and longitudes of palaeomagnetic poles; $A_{95}$: 95 per cent confidence circle radius for poles. 2. Pole 1.15, having doubtful age, is not shown in Fig. 14. 3. *Poles used in the calculations of the APW paths.

**Table 5.** Palaeomagnetic poles obtained for Karelian Province, eastern Fennoscandian Shield—this work.

| Numbers | Rock unit A | NRM component | $_{95}$° | Radiometric age, Ma | Assigned magnetic age, Ma | Polarity | Pole position Φ | ,°N | Λ, °E |
|---|---|---|---|---|---|---|---|---|---|
| 3.5 | Segozero dykes | $B_2$ | | | 1950–1800 | 7N | 46.7 | 261.4 | 14.1 |
| 3.6 | Segozero dykes | $B_1$ | | | 2200–2100 | 5N | −8.3 | 279.7 | 12.3 |
| 3.7 | Segozero dykes | A | | | 2300–2200 | 7R | −41.4 | 311.0 | 13.3 |
| 3.9 | Girvas dolerite dykes | A | | | 2250–2200 | 8R | −30.9 | 280.5 | 13.7 |
| 3.10 | Girvas flows and schists | D | | | 1200 (?) | 6R | −37.2 | 161.5 | 10.7 |
| 3.13 | Girvas flows and schists | $B_2$ | | | 1950–1850 | 6N | 42.2 | 264.7 | 8.8 |
| 2.1 | Matozero dolerites | $B_2$ | | | 1950–1850 | 7N | 43.4 | 256.8 | 8.4 |
| 2.2 | Matozero schists | $B_2$ | | | 1950–1850 | 5N | 39.4 | 269.6 | 11.8 |
| 2.3 | Matozero dolerite dykes | $B_1$ | | 2150 ± 60 | 2200–2100 | 8N | −20.6 | 308.5 | 12.3* |
| 2.4 | Matozero baked schists | $B_1$ | | | 2200–2100 | 8R | −22.1 | 316.8 | 7.9 |
| 2.5 | Matozero dolerite sill | A | | | 2300–2200 | 9R | −38.0 | 294.6 | 9.8 |
| 2.6 | Matozero host schists | A | | | 2300–2200 | 5R | −42.3 | 293.4 | 14.7 |
| 2.9 | Matozero rocks | C | | 2150 ± 60 | 2150–2050 | 10N | 18.2 | 308.4 | 9.6* |
| 2.10 | Shoksha and Sheltozero gabbro-dolerites | $B_2$ | | 1770 ± 12 | 1800–1760 | 6N | 40.5 | 229.8 | 8.1* |
| 2.11 | Shoksha and Sheltozero gabbro | $B_2$ | | 1770 ± 12 | 1780–1760 | 6N | 36.2 | 208.6 | 5.8* |
| 2.12 | Ladva gabbro-dolerites | $B_2$ | | | 1800–1760 | 11N | 40.7 | 218.2 | 5.5 |
| 2.13 | Sheltozero and Ladva rocks | D | | | 1200 (?) | 9R | −39.5 | 168.6 | 8.6 |
| 2.15 | Burakovka gabbro-norites | A | | 2449 ± 1 | 2450–2448 | 16R | −27.6 | 260.0 | 9.2* |
| 2.16 | Burakovka gabbro-diorites | A | | 2439 ± 22 | 2460–2420 | 9R | −25.2 | 247.6 | 10.2 |
| 2.17 | Burakovka ultramafites | A | | | 2400–2300 | 10R | −39.5 | 305.4 | 9.1 |
| 2.18 | Hautavaara gabbro and andesites | A | | 2400 ± 46 | 2450–2350 | 7R | −40.7 | 306.2 | 13.6 |
| 2.19 | Hautavaara gabbro | $B_2$ | | 1880 ± 60 | 1940–1820 | 6N | 32.7 | 272.0 | 9.9* |
| 2.20 | Hautavaara gabbro | D | | 1543 ± 8 | 1550–1535 | 6N | 40.1 | 194.7 | 4.5* |

Notes: 1. Poles 3.6, 3.10, 2.2, 2.6 and 2.13, having doubtful ages or low statistical parameters, are not shown in Fig. 14.2. For other conventions, see notes for Table 4.

also be considered. Indeed, Krasnova & Gooskova (1990) attempted to explain the differences between ChRM directions from the Aganozero and Burakovka blocks of the Burakovka intrusion by the local tectonic movements.

Our new palaeomagnetic data suggest that this interpretation may be correct; however, it is to be noted that the corresponding Burakovka poles (2.15, 2.16 and 2.17, Table 5) are situated within a band formed from the pole sequence in the range 2450–2250 Ma (Fig. 14). Thus, it is also possible to explain the difference in the pole positions of the various blocks of the Burakovka intrusion by the non-coeval ages of the isolated components of NRM. In this case, we follow the interpretation of Mertanen *et al.* (1989) for the NRM components of the layered intrusion in northern Finland. As to the other intrusive bodies investigated, we do not believe that local tectonic movements have affected the palaeomagnetic data to any great extent. Most of these bodies belong to the post-orogenic intrusive events (Sokolov 1987) and, hence, have not tilted significantly after cooling. The subvertical positions of dykes, the roughly horizontal layering of the mafic-ultramafic intrusions and their large lateral dimensions justify this conclusion. Negative results of fold tests, based on the structural attitudes of the layering near the intrusive body boundaries as well as on the foliation of country metamorphic rocks, also indicate this to be true. Because of this we calculate all palaeomagnetic directions for these bodies (plutons, sills and dykes) as well as for the adjacent metamorphosed country rocks in 'geographical coordinates' (i.e. without 'tilt' corrections). Data for the basaltic flows and schists from Girvas are represented in 'stratigraphic coordinates' and are corrected for tilt.

Some new palaeomagnetic results for the eastern Fennoscandian Shield have appeared since Elming *et al.*'s (1993) synthesis. These results are summarized in Table 6, and the associated poles are plotted in Fig. 14, along with our results. The set of 25 reliable ($N > 6$, $A_{95} < 15°$) radiometrically dated poles, which correspond to the B category (Briden & Duff 1981), and the palaeomagnetic directions extracted from Tables 5 and 6, permit us to construct an APWP for the Kola, Belomorian and Karelian provinces, using the spherical spline technique (Jupp & Kent 1987) and its implementation in the GMAP program (Torsvik *et al.* 1990). Fig. 15 demonstrates a good agreement between APWPs for these provinces and their deviations from Elming *et al.*'s (1993) APWP for pre-Svecofennian times. Our new results imply a more complex shape of the pre-Svecofennian part of the Fennoscandian APWP. It is possible to draw a NW–SE track for the APWP, for the interval between 2450 and 2250 Ma, based on the data listed in Tables 5 and 6 (Figs 14 and 15). This track could not have been drawn before now because of the lack of palaeomagnetic data between 2450 and 2120 Ma (Elming *et al.* 1993).

The plate tectonic interpretation of these results remains problematic. The new evidence by no means denies the possibility of constructing a unique APWP for the Fennoscandian Shield, with an accuracy not exceeding the values of $A_{95}$ for individual

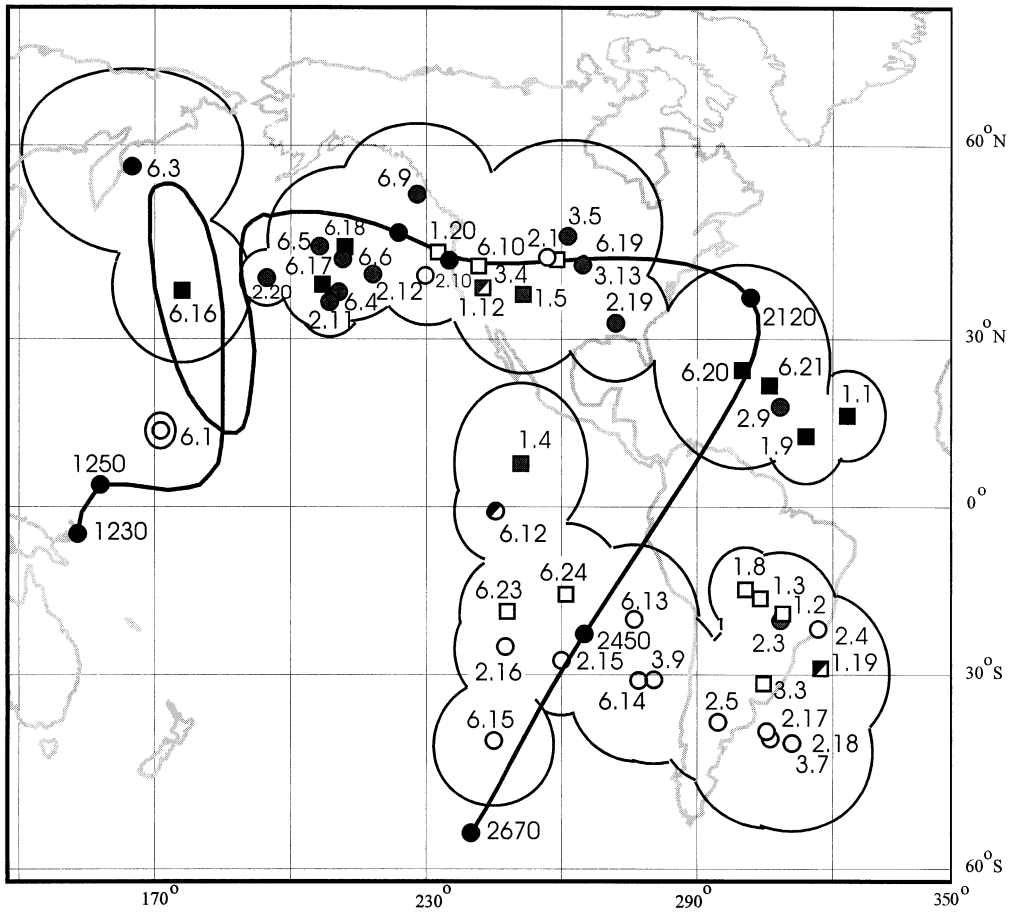



**Figure 14.** The APWP for Fennoscandia for the time interval 2450–1200 Ma (after Elming *et al.* 1993) and the new coeval palaeomagnetic poles. Black dots: key poles after Elming *et al.* (1993). New poles for eastern Fennoscandia are denoted by quadrangles for the Kola and for the Belomorian provinces and by circles for Karelian Province. Black symbols: normal polarity; open symbols: reverse polarity; half-shaded symbols: mixed polarity. Pole numbers are the same as in Tables 4, 5 and 6.

poles (5°–14°). We suggest that no large-scale relative movements have occurred between the various tectonic units of the Fennoscandian Shield since 2400 Ma. This accuracy of 5°–14° for the poles constrains the upper limit of small-scale (up to some hundreds of kilometres and 10 degrees) displacement and block rotations within the shield. More palaeomagnetic studies are needed to reconstruct the relative movements of the individual blocks that make up the Fennoscandian Shield for times prior to this.

On the other hand, our new palaeomagnetic data confirm the considerable latitudinal displacements of the entire Fennoscandian Shield and its orientation change relative to the meridian between 2400 and 1750 Ma. The 'smoothness' of the APWP in this range with a great extent of trajectory looks plausible if we proceed from a hypothesis that the Fennoscandian Shield had then been part of a large lithospheric plate, such as the Proterozoic supercontinent of Piper (1987), and had been moving along with it. The cusps in the APWP near 2300 Ma, 2000 Ma and 1700 Ma (Figs 14 and 15) may be related to large-scale plate tectonic events or re-organizations which changed the kinematics of the whole supercontinent, or the position of the Fennoscandian Shield inside it. The more complex APWP shape for the Shield in post-Svecofennian times with respect to previous ones (Elming *et al.* 1993) may also be related to tectonic events, such as those proposed by Rogers (1996), which relate the collision of the Fennoscandian Shield and Arctica (1500 Ma) to the formation of the Nena continent, and the creation (1000 Ma) and the dispersal (700 Ma) of the Rodinia supercontinent.

Some of these new palaeomagnetic poles might be regarded as reference poles according to the statistical, radiometric, rock-magnetic and structural data. When dealing with the Kola and Belomorian blocks, we cite our pole for the Main Range gabbro (16.7°S, 304.8°E, $A_{95} = 8.3°$; U–Pb age of 2436 ± 25 Ma), our pole for the Voche–Lambina area ultramafic rocks (12.9°N, 314.1E, $A_{95} = 8.6°$; Rb–Sr age of 2198 ± 172 Ma) and our pole for the Kolvitsa peridotites and schists (44.2°N, 232.6°E, $A_{95} = 5.4°$; magnetization age of 1900 Ma) as reference poles. With regard to the Karelian block, we cite our pole for the Burakovka intrusion gabbro-norites (27.6°S, 260.0°E, $A_{95} = 9.2°$; Sm–Nd age of 2439 ± 22 Ma), our pole for the Matozero area dolerites (20.6°S, 308.5°E, $A_{95} = 12.3°$; U–Pb age of 2150 ± 60 Ma), and our pole for the Ropruchey Sill gabbro-dolerites (40.5°N, 229.8°E, $A_{95} = 8.1°$; U–Pb age of 1770 ± 12 Ma) as being reference poles.

The use of these data, as well as the APWPs for the Fennoscandian Shield provinces, in Proterozoic palaeogeographical reconstructions requires a resolution of the polarity

**Table 6.** Some new Early Proterozoic palaeomagnetic poles for the eastern Fennoscandian Shield.

| Numbers | Rock unit | NRM component | Radiometric age, Ma | Assigned magnetic age, Ma | Polarity | Pole position Φ ,°N | | Λ, °E | Reference A $_{95}$° |
|---|---|---|---|---|---|---|---|---|---|
| **Karelian Province (61.5°–66.2°N; 30.2°–36.2°E)** | | | | | | | | | |
| 6.1 | Rybreka sill | B | 1770 ± 12 | 1580 | 7R | 13.9 | 171.3 | 3.2 | Damm *et al.* (1997) |
| 6.2 | Shoksha quartzite | B | 1787 ± 77 | 1580 | 3R | 10.5 | 201.8 | 11.2 | Damm *et al.* (1997) |
| 6.3 | Tikshozero intrusion and dykes | B | 2470 ± 65 | 1700 | 17N | 57.0 | 165.0 | 13.0 | Krasnova & Gooskova (1995) |
| 6.4* | Rybreka sill | A | 1770 ± 12 | 1770 | 7N | 37.8 | 210.6 | 5.5 | Damm *et al.* (1997) |
| 6.5* | Shoksha quartzite. | A | 1787 ± 77 | 1770 | 16N | 45.1 | 206.4 | 4.8 | Damm *et al.* (1997) |
| 6.6 | Girvas lavas. | A | | 1800 | 15N | 43.1 | 211.5 | 6.6 | Damm *et al.* (1997) |
| 6.7 | Lupzenga dykes. | A | 2449 ± 4 | 1840 | 5N | 51.4 | 216.2 | 9.4 | Mertanen (1995) |
| 6.8 | Suopera dykes | A | 2446 ± 5 | 1840 | 5N | 57.1 | 230.6 | 6.1 | Mertanen (1995) |
| 6.9 | Tikshozero intrusion and dykes | A | 2470 ± 65 | 1900 | 13N | 53.0 | 228.0 | 10.0 | Krasnova & Gooskova (1995) |
| 6.10* | Shoksha basalt | | 1810 ± 100 | 1850 | 7N | 42.9 | 235.1 | 8.5 | Damm *et al.* (1997) |
| 6.11 | Shoksha quartzite. | A | 1810 ± 100 | 1850 | 5R | 44.2 | 241.5 | 7.6 | Damm *et al.* (1997) |
| 6.12 | Girvas lavas and quartzite | F | | 2200 | 5N | −0.8 | 245.4 | 9.1 | Damm *et al.* (1997) |
| 6.13* | Lupzenga dykes | D | 2449 ± 4 | 2440 | 13R | −20.4 | 276.1 | 13.5 | Mertanen (1995) |
| 6.14* | Suopera dykes | D | 2446 ± 5 | 2446 | 13R | −31.0 | 277.1 | 13.3 | Mertanen (1995) |
| 6.15* | Tikshozero intrusion and dykes | C | 2470 ± 65 | 2410 | 5R | −41.0 | 245.0 | 10.0 | Krasnova & Gooskova (1995) |
| **Kola and Belomorian Provinces (67.2°–69.5°N; 29.5°–32.6°E)** | | | | | | | | | |
| 6.16 | Pechenga sediments | B | | 1750 | 9N | 38.0 | 176.0 | 12.0 | Goriainov & Turemnov (1996) |
| 6.17 | Pansky layered intrusion gabbro-norites | B | 1850 ± 30** | 1800 | 18N | 39.0 | 207.0 | 7.1 | Dudkin & sion Turemnov (1995) |
| 6.18 | Gabbro-dolerite dykes | B | | 1800 | 7N | 45.0 | 212.0 | 12.0 | Dudkin & Turemnov (1995) |
| 6.19* | Pechenga sediments | A | 1970 ± 5 | 1970 | 16N | 43.0 | 259.0 | 10.0 | Goriainov & Turemnov (1996) |
| 6.20* | Kuetsyarvi Formation | | 2125 ± 50 | 2125 | 10N | 24.7 | 300.8 | 17.6 | Torsvik & Meert (1995) |
| 6.21* | Gabbro-norites | E | 2100 ± 100 | 2150 | 26N | 22.0 | 306.0 | 6.9 | Goriainov & Turemnov (1996) |
| 6.22 | Gabbro-dolerite dykes | E | | 2200 | 3N | 15.0 | 284.0 | 15.7 | Dudkin & dykesTuremnov (1995) |
| 6.23* | Gabbro-norites | D | 2450 ± 50 | 2400 | 31R | −19.0 | 248.0 | 10.0 | Goriainov & Turemnov (1996) |
| 6.24* | Umba Suite | D | 2450 ± 42 | 2400 | 13R | −16.0 | 261.0 | 13.0 | Goriainov & Turemnov (1996) |

Notes: *Poles used in the calculation of the APW paths shown in Fig. 15. For Φ, Λ, $A_{95}$, see notes to Table 4. **K–Ar age of the metamorphic event.

ambiguity (i.e. to decide whether north or south Earth pole positions are used). Following tradition, we suggest as the north poles those Early Proterozoic palaeomagnetic poles that are situated in the present-day western hemisphere, and we assign a normal polarity to the corresponding rock magnetizations (Tables 5 and 6). Recently obtained palaeomagnetic poles have been plotted according to this choice of polarity. However, the latest version of the Vendian–Permian APWP for Baltica (Torsvik *et al.* 1996) implies that the Vendian pole situated in Siberia is the south pole. If this is true, and if the Neoproterozoic track of the APWP (Elming *et al.* 1993) is correct, the Early Proterozoic poles, forming the Early Proterozoic part of the APWP, are also south poles.

Another problem emerges when comparing pole positions for the Early Proterozoic (Fig. 14) or the proposed APWPs (Fig. 15) with the recent versions of the APWP for Vendian–Permian times (e.g. Torsvik *et al.* 1995). It turns out that the 2300–2200 Ma poles lie near the south poles for Devonian times, whereas the 1580–1230 Ma loop, drawn by Elming *et al.* (1993), partly repeats the Devonian to Permian north-pole track for Baltica. Some poles, obtained from different eastern Fennoscandian provinces (poles 1.4, 1.15, 6.12 and 6.23) are situated in the vicinity of the 870–845 Ma part of the APWP of Elming *et al.* (1993). This problem cannot be neglected, because manifestations of younger magmatic activity are well known, for example in the Kola Province, where Caledonian and Hercynian ages for some intrusives have been obtained (Guidebook 1984).

Finally, it is necessary to emphasize, that the time calibrations of the 2300–1900 Ma tracks of the APWPs discussed here are still tentative. Tables 4, 5 and 6 demonstrate the lack of precise radiometric ages in this time interval. Moreover, a new U–Pb zircon age of 1986 ± 18 Ma for the basalt flow which forms the base of the Jatulian sequence in the Girvas

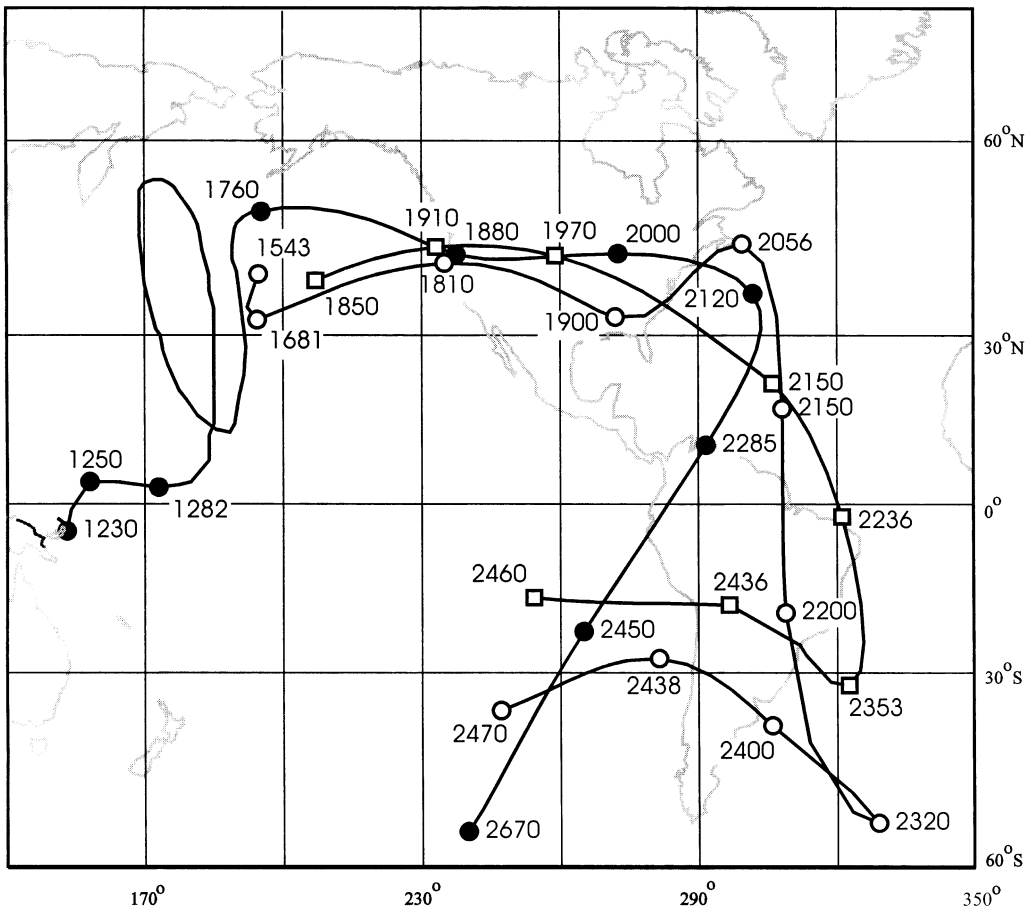



**Figure 15.** A comparison of new APWPs for the Kola–Belomorian (quadrangles) and Karelian (open circles) provinces with that of Elming *et al.* (1993) for Fennoscandia (closed circles). APWPs were calculated using the spherical spline technique of Jupp & Kent (1987).

area (Puchtel *et al.* 1998) contradicts the previous dating of Jatulian rocks. New isotope dating is an obvious necessity in resolving this controversy.

Nevertheless, our new additions to the Proterozoic palaeomagnetic database for the Fennoscandian Shield serve to constrain better the history of its pre-Svecofennian latitudinal displacements and rotations. In order to settle the problem of possible intracratonic movements it will be necessary to obtain more principal palaeomagnetic data for each tectonic unit in the eastern part of the shield, as well as radiometric age evidence for the same geological bodies or bodies that might easily be correlated with them.

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